CLOSED LAYERS IN TENSOR-TRIANGULAR GEOMETRY

PAUL BALMER, RHEA KOMMERELL, AND EMMY VAN ROOY

Abstract. In the absence of a theory of closed immersions in tensor-triangular geometry, we propose an alternative approach via quotients of module categories. As an application, we prove invariance of homological residue fields under completion.

1. Introduction

Let us recall some commutative algebra. Let $I \subseteq R$ be a finitely generated ideal and $\hat{R} = \lim_n R/I^n$ the completion of R along the closed subset $Z = V(I)$ of $\operatorname{Spec}(R)$. Then:

(A) The ring homomorphism $R \to \hat{R}$ induces an isomorphism $R/I \xrightarrow{\sim} \hat{R}/I\hat{R}$.

As a consequence, the spectrum and the residue fields do not change on the locus of completion:

(B) The induced map $\varphi\colon \operatorname{Spec}(\hat{R}) \to \operatorname{Spec}(R)$ restricts to a bijection $\varphi^{-1}(Z) \xrightarrow{\sim} Z$.

(C) For every point $\mathfrak{p} \in Z$, the homomorphism $R \to \hat{R}$ induces an isomorphism between the residue field of R at $\mathfrak{p}$ and the residue field of $\hat{R}$ at the unique point $\hat{\mathfrak{p}}$ in $\varphi^{-1}(\mathfrak{p})$.

The ring $\hat{R}$ and statements (B) and (C) only depend on the closed subset Z. On the other hand, (A) is a somewhat deeper statement since R/I depends on the choice of the ideal I.

Sanders and the first-named author generalized (B) to tensor-triangular geometry as follows (see Recollection 5.20). Let $\mathcal{T}$ be a 'big' tensor triangulated category and $\mathcal{T}^{\mathrm{d}} = \mathcal{T}^{\mathrm{c}}$ the subcategory of dualizable-compact objects. Let $Y \subseteq \operatorname{Spc}(\mathcal{T}^{\mathrm{d}})$ be a Thomason subset of the tt-spectrum, *e. g.* a closed subset with quasi-compact complement. Consider the 'Y-completion' functor $\mathrm{Compl}_Y = (-)^{\wedge}_Y = [\mathbb{e}_Y, -]$ and the induced map on spectra

$$\mathrm{Compl}_Y\colon \mathcal{T}^{\mathrm{d}} \to \widehat{\mathcal{T}}^{\mathrm{d}} \qquad \text{and} \qquad \varphi = \operatorname{Spc}(\mathrm{Compl}_Y)\colon \operatorname{Spc}(\widehat{\mathcal{T}}^{\mathrm{d}}) \to \operatorname{Spc}(\mathcal{T}^{\mathrm{d}}). \tag{1.1}$$

Then φ restricts to a bijection $\varphi^{-1}(Y) \xrightarrow{\sim} Y$. This is [BS25, Theorem 4.6], and the same holds for homological spectra (Corollary 5.14). These results generalize (B) to tt-geometry.

It is much harder to generalize (A), mainly because we lack a tensor-triangular analog of the quotient R/I; see Remark 2.15. Case in point, the notion of 'residue field' led to ruminations [BKS19] and unnerving [BHR26, Cou26]. In this paper, we propose a workaround that allows us in particular to generalize (C) to tensor-triangular geometry:

1.2. **Theorem** (Corollary 5.21)**.** *Let $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^{\mathrm{d}})$ be a homological prime on Y and $\hat{\mathcal{H}} \in \operatorname{Spc}^{\mathrm{h}}(\widehat{\mathcal{T}}^{\mathrm{d}})$ its unique preimage. Then the Y-completion functor (1.1) induces an equivalence*

$$\frac{\text{Mod-}\mathcal{T}^{\mathrm{d}}}{\mathrm{Loc}(\mathcal{H})} \xrightarrow{\ \sim\ } \frac{\text{Mod-}\widehat{\mathcal{T}}^{\mathrm{d}}}{\mathrm{Loc}(\hat{\mathcal{H}})} \tag{1.3}$$

of the corresponding homological residue fields.

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We discuss completion mostly as an illustration. Our higher ambition is to understand the correct tensor-triangular analog of closed subschemes. As this remains elusive, we propose an indirect approach via what we call *layers* (Definition 2.10), *i.e.* finite quotients of the module category $\mathcal{A} = \text{Mod-}\mathcal{K}$ over any rigid tensor triangulated category $\mathcal{K}$. More precisely, a layer

$$\bar{\mathcal{A}}_{\mathcal{I}} \coloneqq \frac{\text{Mod-}\mathcal{K}}{\text{Loc}\,(\mathcal{I})} \tag{1.4}$$

is the Gabriel quotient of $\mathcal{A}$ by a Serre $\otimes$-ideal $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$ of finitely presented $\mathcal{K}$-modules.

Although loosely inspired by the quotients R/I of commutative algebra, the above notion of a layer is actually much broader. For instance, homological residue fields (1.3) are layers – the *minimal* ones. More strikingly, for every quasi-compact *open* subset $U \subseteq \text{Spc}(\mathcal{K})$, the module category $\text{Mod-}\mathcal{K}(U)$ of the open piece $\mathcal{K}(U)$ is also a layer as in (1.4); see [Bal20b, Proposition 2.9]. To understand what the *closed* layers are, let us discuss some spectra.

1.5. *Hypothesis.* To keep things simple, assume that the homological spectrum and the tt-spectrum coincide: $\text{Spc}^{\text{h}}(\mathcal{K}) = \text{Spc}(\mathcal{K})$. We drop this assumption after the Introduction.

Every layer $\bar{\mathcal{A}}_{\mathcal{I}}$ as in (1.4) has a natural 'spectrum' (Definition 3.1) that identifies with the following subset of the spectrum of $\mathcal{K}$:

$$W(\mathcal{I}) = \big\{\mathcal{H} \in \text{Spc}^{\text{h}}(\mathcal{K}) \,\big|\, \mathcal{I} \subseteq \mathcal{H}\big\}. \tag{1.6}$$

The notation W is meant to evoke the closed subset $V(I) = \big\{\mathfrak{p} \in \text{Spec}(R) \,\big|\, I \subseteq \mathfrak{p}\big\}$ of commutative algebra while reminding us that we are in a broader setting: $W(\mathcal{I})$ need not be a closed subset in general. Here are the actual possibilities.

1.7. **Theorem** (Section 3)**.** *The spectrum $W(\mathcal{I})$ is a proconstructible subset (Recollection 3.5) of* $\text{Spc}(\mathcal{K})$*. Conversely, every proconstructible subset is the spectrum of some layer.*

Proconstructible subsets include singletons and quasi-compact open subsets, as illustrated above by the residual layer $\bar{\mathcal{A}}_{\mathcal{H}}$ (whose spectrum is just $\{\mathcal{H}\}$) and the open layer $\text{Mod-}\mathcal{K}(U)$ (whose spectrum is U). Closed layers are simply those layers whose spectrum is closed.

In commutative algebra, several ideals $I \subseteq R$ define the same closed subset $Z = V(I)$. Among them, the radical $\sqrt{I} = \cap_{\mathfrak{p}\in Z}\,\mathfrak{p}$ is the largest. Similarly, many of our Serre $\otimes$-ideals $\mathcal{I}$ define the same proconstructible subset $Y = W(\mathcal{I})$ of $\text{Spc}(\mathcal{K})$. Among them, the maximal one is given by $\mathcal{I}_{\max}(Y) = \cap_{\mathcal{H}\in Y}\,\mathcal{H}$; see Proposition 3.21.

Remarkably, there is also a 'lower bound' when Y is a Thomason subset (see Section 4). This fact does not seem to have an analog in commutative algebra. It is expressed in terms of 'large' subcategories of $\text{Mod-}\mathcal{K}$. Define $\mathcal{B}_{\min}(Y)$ to be the kernel of the functor

$$\mathcal{A} = \text{Mod-}\mathcal{K} \twoheadrightarrow \text{Mod-}\mathcal{K}_Y$$

given by restriction from $\mathcal{K}$ to the tt-ideal $\mathcal{K}_Y = \big\{c \in \mathcal{K} \,\big|\, \text{supp}(c) \subseteq Y\big\}$ of objects supported on Y. This localizing subcategory $\mathcal{B}_{\min}(Y)$ is typically *not* of finite type. (See Proposition 4.4.) We show that $\mathcal{B}_{\min}(Y)$ is a $\otimes$-ideal, which implies that $\text{Mod-}\mathcal{K}_Y$ inherits a tensor product.

1.8. **Theorem** (Theorem 4.6)**.** *Let $Y \subseteq \text{Spc}(\mathcal{K})$ be a Thomason subset and let $\mathcal{B}_{\min}(Y)$ be the above localizing $\otimes$-ideal of $\mathcal{A}$. Let $\mathcal{I}$ be a Serre $\otimes$-ideal of $\mathcal{A}^{\text{fp}} = \text{mod-}\mathcal{K}$. Then the spectrum $W(\mathcal{I})$ of the layer $\bar{\mathcal{A}}_{\mathcal{I}}$ is contained in Y if and only if*

$$\mathcal{B}_{\min}(Y) \subseteq \text{Loc}(\mathcal{I}).$$

In that case, the layer $\bar{\mathcal{A}}_{\mathcal{I}}$ is a Gabriel quotient of $\text{Mod-}\mathcal{K}_Y$ *by a localizing $\otimes$-ideal.*

It follows that the layers that live 'over Y' only depend on the tt-ideal $\mathcal{K}_Y$. To see this, consider $F\colon \mathcal{K} \to \mathcal{L}$ a tt-functor and $\varphi = \operatorname{Spc}(F)\colon \operatorname{Spc}(\mathcal{L}) \to \operatorname{Spc}(\mathcal{K})$ the associated map. For a Thomason subset $Y \subseteq \operatorname{Spc}(\mathcal{K})$, we say that F is an *equivalence over* Y if F restricts to an equivalence $\mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{\varphi^{-1}(Y)}$. For example, the Y-completion functor $\mathcal{T}^{\mathrm{d}} \to \widehat{\mathcal{T}}^{\mathrm{d}}$ of (1.1) is an equivalence over Y, by [BS25, Theorem 4.6]. We can now state our generalization of (A) to tensor-triangular geometry; this occupies our central Section 5.

1.9. **Theorem** (Theorem 5.9)**.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over a Thomason subset Y. Then F induces a bijection between the Serre $\otimes$-ideals $\mathcal{I}$ of* mod-$\mathcal{K}$ *whose spectrum $W(\mathcal{I})$ is contained in Y and the Serre $\otimes$-ideals $\mathcal{J}$ of* mod-$\mathcal{L}$ *whose spectrum $W(\mathcal{J})$ is contained in $\varphi^{-1}(Y)$. Moreover if $\mathcal{I}$ corresponds to $\mathcal{J}$ under this bijection, then F yields an equivalence of the associated layers*

$$\frac{\text{Mod-}\mathcal{K}}{\operatorname{Loc}(\mathcal{I})} \xrightarrow{\sim} \frac{\text{Mod-}\mathcal{L}}{\operatorname{Loc}(\mathcal{J})}.$$

The proof of this result relies on the following characterization of finite type $\otimes$-localizations of Mod-$\mathcal{K}$, that could be of independent interest.

1.10. **Theorem** (Theorem 2.14)**.** *Let $\mathcal{B}$ be a localizing $\otimes$-ideal of $\mathcal{A} =$* Mod-$\mathcal{K}$*. Then $\mathcal{B}$ is generated by $\mathcal{B} \cap \mathcal{A}^{\mathrm{fp}}$ if and only if the $\otimes$-unit $\bar{\mathbb{1}}$ in $\mathcal{A}/\mathcal{B}$ is finitely presented.*

As an illustration of our methods, we show in Section 6 how residual regularity at points of Y (in the sense of [VR26]) is preserved along a tt-functor that is an equivalence over Y. As indicated above, this applies in particular to Y-completion $\mathcal{T}^{\mathrm{d}} \to \widehat{\mathcal{T}}^{\mathrm{d}}$.

1.11. **Theorem** (Theorem 6.7)**.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over a Thomason subset Y. Let $\varphi = \operatorname{Spc}(F)$. Let $\mathcal{Q} \in \operatorname{Spc}(\mathcal{L})$ such that $\mathcal{P} \coloneqq \varphi(\mathcal{Q})$ belongs to Y. Then $\mathcal{L}$ is residually regular at $\mathcal{Q}$ if and only if $\mathcal{K}$ is residually regular at $\mathcal{P}$.*

In the final Section 7, we consider the example of $\mathcal{K} = \mathrm{D}^{\mathrm{perf}}(R)$ for a commutative ring R. We show that the set of ideals of R injects into the set of Serre $\otimes$-ideals of mod-$\mathcal{K}$ in a way that is compatible with vanishing loci, further justifying the notation $W(\mathcal{I})$. This injection fails to be surjective already for the ring $R = \mathbb{Z}$ of integers.

AI Statement. The paper has been written entirely by the authors. All new ideas and all new proofs are their own.

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2. Preliminaries and notation

2.1. *Hypothesis.* Throughout the paper, $\mathcal{K}$ denotes an essentially small tensor triangulated category that we assume idempotent-complete and rigid. Typically, $\mathcal{K}$ is the subcategory of compact objects (or equivalently, dualizable objects) in a 'big' *rigidly-compactly generated* tensor triangulated category $\mathcal{T}$ in the sense of [BF11]:

$$\mathcal{K} = \mathcal{T}^{\mathrm{c}} = \mathcal{T}^{\mathrm{d}}. \tag{2.2}$$

Such 'big' tt-categories $\mathcal{T}$ are extremely common in topology, algebra and geometry. They are called 'unital algebraic stable homotopy categories' in [HPS97], where the reader can find a multitude of examples. From the perspective of applications, there is no restriction in assuming the existence of a 'big' $\mathcal{T}$ satisfying (2.2), as $\mathcal{K}$ can be Ind-completed in the presence of a model. We use $\mathcal{K}$ to emphasize that our theory only depends on small objects.

2.3. *Recollection.* Rigidity of $\mathcal{K}$ implies that every object $c \in \mathcal{K}$ admits a dual $c^\vee \in \mathcal{K}$ with an adjunction $(c\otimes -) \dashv (c^\vee \otimes -)$. Its unit is denoted $\mathrm{coev}_c \colon \mathbb{1} \to c^\vee \otimes c$. As $c \otimes \mathrm{coev}_c \colon c \to c \otimes c^\vee \otimes c$ is a split monomorphism, every object c is a direct summand of $c \otimes c^\vee \otimes c$.

2.4. *Recollection.* See [BKS19, § 2]. We denote the category of right $\mathcal{K}$-modules by

$$\mathcal{A} = \text{Mod-}\mathcal{K} = \mathrm{Add}(\mathcal{K}^{\mathrm{op}}, \mathrm{Ab}).$$

It is a locally coherent Grothendieck abelian category. Its abelian subcategory of finitely presented objects $\mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{K}$, also known as the *Freyd envelope* of $\mathcal{K}$ ([Nee01, Chapter 5]), is Frobenius and its projective-injectives are the representables $\mathrm{h}(c)$ for $c \in \mathcal{K}$, where

$$\mathrm{h} \colon \mathcal{K} \hookrightarrow \text{mod-}\mathcal{K} \hookrightarrow \text{Mod-}\mathcal{K} = \mathcal{A}.$$

is the Yoneda embedding. By Day convolution, $\mathcal{A}$ inherits a tensor, cocontinuous in each variable and making h a tensor functor. The category $\mathcal{A}$ has enough injectives and

$$\textit{every injective in } \mathcal{A} \textit{ is } \otimes\textit{-flat.} \tag{2.5}$$

See [BW26, §2], or [Kra00, Corollary 1.9] and [BKS19, Proposition A.14] when (2.2) holds.

2.6. *Recollection.* Given $\mathcal{K}$ and $\mathcal{L}$ two essentially small triangulated categories, any exact functor $F \colon \mathcal{K} \to \mathcal{L}$ induces a functor $F_! \colon \text{Mod-}\mathcal{K} \to \text{Mod-}\mathcal{L}$, given by left Kan extension

$$F_!(M) = \operatorname*{colim}_{c\in\mathcal{K},\ \mathrm{h}(c)\to M} \mathrm{h}(F(c)).$$

In particular, $F_! \circ \mathrm{h} = \mathrm{h} \circ F$. This functor has a right adjoint $F^* \colon \text{Mod-}\mathcal{L} \to \text{Mod-}\mathcal{K}$, given by restriction along F. Both $F_!$ and F^* are exact cocontinuous functors and $F_!$ preserves finitely presented objects. See [Kra05, §1]. When $\mathcal{K}$ and $\mathcal{L}$ are moreover tt-categories as in Hypothesis 2.1 and F is a tt-functor, then $F_!$ is symmetric monoidal as well, and the adjunction $F_! \dashv F^*$ satisfies a projection formula: $F^*(N \otimes F_!(M)) \cong F^*(N) \otimes M$.

2.7. *Recollection.* The *homological spectrum of* $\mathcal{K}$ is defined as $\mathrm{Spc}^{\mathrm{h}}(\mathcal{K}) = \mathrm{Max}_{\otimes}^{\mathrm{Serre}}(\mathcal{A}^{\mathrm{fp}})$, where $\mathrm{Max}_{\otimes}^{\mathrm{Serre}}$ denotes the collection of maximal Serre $\otimes$-ideals. The latter are called *homological primes*. See [Bal20b]. There is a well-defined surjection

$$\pi \colon \mathrm{Spc}^{\mathrm{h}}(\mathcal{K}) \twoheadrightarrow \mathrm{Spc}(\mathcal{K})$$

sending a homological prime $\mathcal{H} \subset \text{mod-}\mathcal{K}$ to $\pi(\mathcal{H}) = \mathrm{h}^{-1}(\mathcal{H}) \subset \mathcal{K}$. The surjection π is very often bijective, see [Bal20b, § 5]. However, π can fail to be injective, as observed by [BHR26, Cou26]. One sometimes says that $\mathcal{K}$ has the *steel property* if π is a bijection.

2.8. *Recollection.* See [BKS19, § 2]. For every Serre $\otimes$-ideal $\mathcal{I} \subseteq \mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{K}$ we consider the Gabriel quotient by the localizing subcategory generated by $\mathcal{I}$ (which is always $\otimes$-ideal)

$$\begin{array}{c} \mathcal{A} = \text{Mod-}\mathcal{K} \\ {\scriptstyle Q_{\mathcal{I}}}\big\downarrow \dashv \big\uparrow{\scriptstyle R_{\mathcal{I}}} \\ \bar{\mathcal{A}}_{\mathcal{I}} \coloneqq \frac{\text{Mod-}\mathcal{K}}{\mathrm{Loc}(\mathcal{I})} \end{array} \tag{2.9}$$

The Grothendieck category $\bar{\mathcal{A}}_{\mathcal{I}}$ remains locally coherent and its subcategory of finitely presented objects $\bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}}$ identifies with $(\text{mod-}\mathcal{K})/\mathcal{I}$, the quotient of $\mathcal{A}^{\mathrm{fp}}$ by the chosen $\mathcal{I}$. In (2.9), the functor $R_{\mathcal{I}}$ is the fully faithful right adjoint to the exact $\otimes$-functor $Q_{\mathcal{I}}$. The adjunction $Q_{\mathcal{I}} \dashv R_{\mathcal{I}}$ satisfies a projection formula: $R_{\mathcal{I}}(N \otimes Q_{\mathcal{I}}(M)) \cong R_{\mathcal{I}}(N) \otimes M$. We often write $\bar{x}$ instead of $Q_{\mathcal{I}}(x)$ for $x \in \mathcal{A}$, or instead of $Q_{\mathcal{I}}(\mathrm{h}(x))$ for $x \in \mathcal{K}$. Generalizing (2.9), given two Serre $\otimes$-ideals $\mathcal{I} \subseteq \mathcal{H}$, we have an intermediate $Q_{\mathcal{H},\mathcal{I}} \colon \bar{\mathcal{A}}_{\mathcal{I}} \twoheadrightarrow \bar{\mathcal{A}}_{\mathcal{H}}$ such that $Q_{\mathcal{H}} \cong Q_{\mathcal{H},\mathcal{I}} \circ Q_{\mathcal{I}}$.

2.10. **Definition.** We call the above finite Gabriel quotient $\bar{\mathcal{A}}_{\mathcal{I}} = \mathcal{A}/\operatorname{Loc}(\mathcal{I})$ the *layer of* $\mathcal{K}$ *with respect to* $\mathcal{I}$, or the $\mathcal{I}$*-layer of* $\mathcal{K}$ for short.

2.11. *Recollection.* The injective hull of the unit $\bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{I}}$ in $\bar{\mathcal{A}}_{\mathcal{I}}$ yields an injective object $E_{\mathcal{I}} \coloneqq R_{\mathcal{I}}(\bar{E}_{\mathcal{I}})$ in $\mathcal{A} = \text{Mod-}\mathcal{K}$ and a morphism $\eta \colon \mathbb{1} \to E_{\mathcal{I}}$. Since $Q_{\mathcal{I}} \circ R_{\mathcal{I}} = \mathrm{Id}$ we have $Q_{\mathcal{I}}(E_{\mathcal{I}}) = \bar{E}_{\mathcal{I}}$, justifying the notation. (When (2.2) holds, this injective $E_{\mathcal{I}} \in \mathcal{A}$ is the image of a pure-injective in $\mathcal{T}$ also denoted $E_{\mathcal{I}}$ in [BKS19].) Thanks to (2.5), the object $E_{\mathcal{I}}$ is $\otimes$-flat in $\mathcal{A}$ and it follows that it is a weak ring ($E_{\mathcal{I}} \otimes \eta \colon E_{\mathcal{I}} \to E_{\mathcal{I}} \otimes E_{\mathcal{I}}$ has a retraction). We have

$$\operatorname{Loc}(\mathcal{I}) = \operatorname{Ker}(E_{\mathcal{I}} \otimes - \colon \mathcal{A} \to \mathcal{A}) \quad \text{and} \quad \mathcal{I} = \operatorname{Ker}(E_{\mathcal{I}} \otimes -) \cap \mathcal{A}^{\mathrm{fp}}. \tag{2.12}$$

This is proved under (2.2) in [BKS19] but the proof works in general; see [BW26, § 2].

2.13. **Notation.** We follow [VR26] and depart from earlier literature by writing

$$\kappa(\mathcal{H})$$

for the injective object $E_{\mathcal{I}}$ when $\mathcal{I} = \mathcal{H}$ is a homological prime $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. This is justified by [BC21] where these objects are computed to be usual residue fields $\kappa(x)$ in algebraic geometry (not their injective hulls!) and to be Morava K-theories $K(n)$ in topology.

We believe that the following result is new. The 'if' direction is somewhat surprising: One can decide whether a localizing $\otimes$-ideal $\mathcal{B}$ is of finite type solely in terms of $\mathcal{A}/\mathcal{B}$.

2.14. **Theorem.** *Let $\mathcal{B}$ be a localizing $\otimes$-ideal subcategory of $\mathcal{A} = \text{Mod-}\mathcal{K}$. Then $\mathcal{B}$ is of finite type, i.e. $\mathcal{B} = \operatorname{Loc}(\mathcal{B} \cap \mathcal{A}^{\mathrm{fp}})$, if and only if the $\otimes$-unit $\bar{\mathbb{1}}$ in $\mathcal{A}/\mathcal{B}$ is finitely presented.*

Proof. The forward direction is classical [Kra97, Theorem 2.6]: $\mathcal{A}/\mathcal{B}$ remains locally coherent and $Q_{\mathcal{B}} \colon \mathcal{A} \twoheadrightarrow \mathcal{A}/\mathcal{B}$ preserves finitely presented objects, including here $\mathbb{1}_{\mathcal{A}} = \mathrm{h}(\mathbb{1}_{\mathcal{K}}) \in \text{mod-}\mathcal{K} = \mathcal{A}^{\mathrm{fp}}$. The reverse direction is the interesting one. It suffices to prove that $Q_{\mathcal{B}}$ preserves finitely presented objects. Indeed, the usual argument using the adjunction $Q_{\mathcal{B}} \dashv R_{\mathcal{B}}$ and the fact that $\mathcal{A}$ is locally finitely presented shows that $R_{\mathcal{B}} \colon \mathcal{A}/\mathcal{B} \to \mathcal{A}$ preserves filtered colimits, and this is sufficient to guarantee that $\mathcal{B}$ is of finite type by [Kra97, Lemma 2.4]. The little miracle here is that $Q_{\mathcal{B}}$, being a tensor-functor, preserves dualizable objects. It is easy to show in general that if $\bar{\mathbb{1}}$ is finitely presented in $\mathcal{A}/\mathcal{B}$ then any dualizable d is finitely presented as well, *i.e.* $\operatorname{Hom}(d, -)$ preserves filtered colimits: use $\operatorname{Hom}(d, -) \cong \operatorname{Hom}(\bar{\mathbb{1}}, d^{\vee} \otimes -)$ and the fact that $d^{\vee} \otimes -$ commutes with all colimits. Now in $\mathcal{A}$ every finitely presented object is a cokernel of a map between dualizable ones, and $Q_{\mathcal{B}}$ is exact. The result follows easily since a cokernel of a morphism of finitely presented objects remains finitely presented. $\square$

2.15. *Remark.* The concept of 'closed immersion' has not been extended to tensor-triangular geometry, in the sense that there is no purely tensor-triangular construction that would produce the derived category $\mathrm{D}(Z)$ of a closed subscheme $Z \hookrightarrow X$ when applied to $\mathcal{T} = \mathrm{D}(X)$. In particular, the naive guess $\mathcal{T}_Z$ produces $\mathrm{D}_Z(X)$, which only depends on the closed *subset* Z, as opposed to $\mathrm{D}(Z)$ that depends on the sheaf of ideals defining Z. In fact, no closed subscheme of X can have $\mathrm{D}_Z(X)$ as its derived category because $\mathrm{D}_Z(X)$ is not rigidly-compactly generated except in trivial cases. See a detailed discussion in [BS25, Remark 2.12].

2.16. *Recollection.* Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a Thomason subset and consider the tt-ideal $\mathcal{K}_Y = \{a \in \mathcal{K} \mid \operatorname{supp}(a) \subseteq Y\}$. When (2.2) holds, we can construct in $\mathcal{T}$ the idempotents $\mathbb{e}_Y$ and $\mathbb{f}_Y$ of [BF11]. We explain how their images in $\mathcal{A} = \text{Mod-}\mathcal{K}$ can be constructed in the absence of $\mathcal{T}$. Recollection 2.6 for the inclusion $i\colon \mathcal{K}_Y \hookrightarrow \mathcal{K}$ and the quotient $q\colon \mathcal{K} \twoheadrightarrow \mathcal{K}/\mathcal{K}_Y$ gives

$$\begin{array}{ccc} \text{Mod-}\mathcal{K}_Y & & \text{Mod-}\mathcal{K} \\ {\scriptstyle i_!}\downarrow \dashv \uparrow{\scriptstyle i^*} & & {\scriptstyle q_!}\downarrow \dashv \uparrow{\scriptstyle q^*} \\ \text{Mod-}\mathcal{K} & & \text{Mod-}\mathcal{K}/\mathcal{K}_Y \end{array}$$

Using these adjunctions, we may define the following objects in Mod-$\mathcal{K}$:

$$\mathbb{e}_Y \coloneqq i_! i^*(\mathbb{1}) \qquad \text{and} \qquad \mathbb{f}_Y \coloneqq q^* q_!(\mathbb{1}).$$

The explicit formulas for these adjunctions give

$$\mathbb{e}_Y = \operatorname*{colim}_{\substack{a \to \mathbb{1} \text{ in } \mathcal{K} \\ \text{with } a \in \mathcal{K}_Y}} \mathrm{h}(a) \qquad \text{and} \qquad \mathbb{f}_Y = \operatorname{Hom}_{\mathcal{K}/\mathcal{K}_Y}(q(-), \mathbb{1}) = \operatorname*{colim}_{\substack{\sigma\colon \mathbb{1} \to b \text{ in } \mathcal{K} \\ \text{with } \operatorname{cone}(\sigma) \in \mathcal{K}_Y}} \mathrm{h}(b). \tag{2.17}$$

Since $\mathbb{e}_Y$ and $\mathbb{f}_Y$ are filtered colimits of $\otimes$-flat objects, they are themselves $\otimes$-flat. By [BIK15, Proposition 2.10], the counit $\mathbb{e}_Y \to \mathbb{1}$ and unit $\mathbb{1} \to \mathbb{f}_Y$ fit into a long exact sequence in $\mathcal{A}$

$$\cdots \to \Sigma^{-1}\mathbb{f}_Y \to \mathbb{e}_Y \to \mathbb{1} \to \mathbb{f}_Y \to \Sigma\mathbb{e}_Y \to \dots \tag{2.18}$$

Using the projection formula for the adjunction $q_! \dashv q^*$ (Recollection 2.6), we see that

$$\mathbb{f}_Y \otimes \mathrm{h}(a) \cong q^*(q_!(\mathrm{h}(a))) \cong q^*(\mathrm{h}(q(a))) = 0 \qquad \text{for all } a \in \mathcal{K}_Y. \tag{2.19}$$

This gives $\mathbb{e}_Y \otimes \mathbb{f}_Y = 0$ by (2.17). Hence $\mathbb{e}_Y \otimes \mathbb{e}_Y \xrightarrow{\sim} \mathbb{e}_Y$ and $\mathbb{f}_Y \xrightarrow{\sim} \mathbb{f}_Y \otimes \mathbb{f}_Y$ by (2.18) and $\otimes$-flatness of $\mathbb{e}_Y$ and $\mathbb{f}_Y$. So $\mathbb{e}_Y$ is a $\otimes$-idempotent co-ring and $\mathbb{f}_Y$ a $\otimes$-idempotent ring in $\mathcal{A}$.

2.20. **Lemma** (Compare [BHS23, Lemma 3.8])**.** *With notation as in Recollection 2.16.*

(a) *We have* $\big\{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,\big|\, Q_{\mathcal{H}}(\mathbb{f}_Y) = 0\big\} = \big\{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,\big|\, Q_{\mathcal{H}}(\mathbb{e}_Y) = \bar{\mathbb{1}}\big\} = \pi^{-1}(Y)$.

(b) *We have* $\big\{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,\big|\, Q_{\mathcal{H}}(\mathbb{f}_Y) = \bar{\mathbb{1}}\big\} = \big\{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,\big|\, Q_{\mathcal{H}}(\mathbb{e}_Y) = 0\big\} = \pi^{-1}(Y)^{\mathfrak{C}}$.

Proof. There are exactly two cases: If $\mathcal{H} \in \pi^{-1}(Y)$ then there exists an object $a \in \mathcal{K}_Y$ with $\pi(\mathcal{H}) \in \operatorname{supp}(a)$, meaning $\bar{a} \neq 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$. But $\overline{\mathbb{f}_Y} \otimes \bar{a} = 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$ because of (2.19), forcing $\overline{\mathbb{f}_Y} = 0$. This gives $\overline{\mathbb{e}_Y} \cong \bar{\mathbb{1}}$ by (2.18). The other case is $\mathcal{H} \in \pi^{-1}(Y)^{\mathfrak{C}}$. Then $\pi(\mathcal{H}) \notin \operatorname{supp}(a)$ for all $a \in \mathcal{K}_Y$. Hence $\bar{a} = 0$ for all $a \in \mathcal{K}_Y$ and it follows from (2.17) that $\overline{\mathbb{e}_Y} = 0$ as well. Again, (2.18) guarantees that $\overline{\mathbb{f}_Y} \cong \bar{\mathbb{1}}$ in this case. This gives the result. □

3. Spectra of layers

We keep $\mathcal{K}$ as in Hypothesis 2.1 and extend the definition of the homological spectrum to every layer. We characterize the corresponding subsets of $\operatorname{Spc}(\mathcal{K})$ topologically.

3.1. **Definition.** By extension of Recollection 2.7, the *homological spectrum of the layer* $\bar{\mathcal{A}}_{\mathcal{I}}$ (see Definition 2.10) refers to the set $\operatorname{Max}_{\otimes}^{\mathrm{Serre}}(\bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}})$ of maximal Serre $\otimes$-ideals of $\bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}}$. Since $\mathcal{A}^{\mathrm{fp}} \twoheadrightarrow \bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}}$ is the quotient by the chosen Serre $\otimes$-ideal $\mathcal{I}$, we have an injection $\operatorname{Max}_{\otimes}^{\mathrm{Serre}}(\bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}}) \hookrightarrow \operatorname{Max}_{\otimes}^{\mathrm{Serre}}(\mathcal{A}^{\mathrm{fp}}) = \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ given by $\bar{\mathcal{H}} \mapsto Q_{\mathcal{I}}^{-1}(\bar{\mathcal{H}})$. This identifies $\operatorname{Max}_{\otimes}^{\mathrm{Serre}}(\bar{\mathcal{A}}_{\mathcal{I}}^{\mathrm{fp}})$ with the following subset of $\operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$, that we also call the *homological spectrum of* $\bar{\mathcal{A}}_{\mathcal{I}}$:

$$W^{\mathrm{h}}(\mathcal{I}) \coloneqq \big\{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,\big|\, \mathcal{I} \subseteq \mathcal{H}\big\}.$$

Its image under $\pi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \twoheadrightarrow \operatorname{Spc}(\mathcal{K})$ is *the spectrum of* $\bar{\mathcal{A}}_{\mathcal{I}}$

$$W(\mathcal{I}) \coloneqq \pi(W^{\mathrm{h}}(\mathcal{I})).$$

So the subset $W(\mathcal{I}) \subseteq \operatorname{Spc}(\mathcal{K})$ consists of the primes $\mathcal{P} \in \operatorname{Spc}(\mathcal{K})$ where *there exists* a homological prime $\mathcal{H}$ above $\mathcal{P}$ such that $\mathcal{I} \subseteq \mathcal{H}$, but there is no guarantee that this holds for all $\mathcal{H} \in \pi^{-1}(\mathcal{P})$. When $\mathcal{K}$ has the steel property, this distinction is irrelevant.

3.2. **Lemma.** *Let $\mathcal{I} \subseteq \operatorname{mod}\text{-}\mathcal{K}$ be a Serre $\otimes$-ideal and let $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. Then $\mathcal{H}$ belongs to the homological spectrum $W^{\mathrm{h}}(\mathcal{I})$ of the $\mathcal{I}$-layer $\bar{\mathcal{A}}_{\mathcal{I}}$ if and only if $\kappa(\mathcal{H}) \otimes E_{\mathcal{I}} \neq 0$ in $\operatorname{Mod}\text{-}\mathcal{K}$, that is, by (2.12), if and only if $\kappa(\mathcal{H})$ does not belong to $\operatorname{Loc}(\mathcal{I})$.*

Proof. If $\mathcal{H}$ belongs to $W^{\mathrm{h}}(\mathcal{I})$, then $\mathcal{I} \subseteq \mathcal{H}$ and the Gabriel quotient $Q_{\mathcal{H}}$ factors via $\bar{\mathcal{A}}_{\mathcal{I}}$ as $Q_{\mathcal{H}} \cong Q_{\mathcal{H},\mathcal{I}} \circ Q_{\mathcal{I}}$. Since $Q_{\mathcal{H}}(\kappa(\mathcal{H})) = \bar{E}_{\mathcal{H}} \neq 0$ this forces $Q_{\mathcal{I}}(\kappa(\mathcal{H})) \neq 0$, *i.e.* $\kappa(\mathcal{H}) \notin \operatorname{Loc}(\mathcal{I})$.

Conversely, suppose that $\kappa(\mathcal{H}) \otimes E_{\mathcal{I}} \neq 0$. Then, $\mathcal{H}' \coloneqq \operatorname{Ker}(\kappa(\mathcal{H}) \otimes E_{\mathcal{I}} \otimes -) \cap \mathcal{A}^{\mathrm{fp}}$ is a proper Serre $\otimes$-ideal of mod-$\mathcal{K}$ containing both $\mathcal{I}$ and $\mathcal{H}$, which is maximal. Hence $\mathcal{H} = \mathcal{H}' \supseteq \mathcal{I}$. □

3.3. *Recollection.* Let $f\colon \mathbb{1} \to X$ be a morphism in $\mathcal{A}$ with X $\otimes$-flat. The Nilpotence Theorem [Bal20a, Corollary 4.2] says that if $\bar{f} = 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$ for all $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ then $f^{\otimes n} = 0$ for $n \gg 1$. If f goes to zero in $\bar{\mathcal{A}}_{\mathcal{H}}$ only for those $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$ then f is $\otimes$-nilpotent in $\bar{\mathcal{A}}_{\mathcal{I}}$. One can either adapt the proof in *loc. cit.* to $\bar{\mathcal{A}}_{\mathcal{I}}$, or apply the original Nilpotence Theorem to $\eta \otimes f\colon \mathbb{1} \to E_{\mathcal{I}} \otimes X$ in $\mathcal{A}$, which vanishes also in $\bar{\mathcal{A}}_{\mathcal{H}}$ for $\mathcal{H} \notin W^{\mathrm{h}}(\mathcal{I})$ by Lemma 3.2.

3.4. *Remark.* The spectrum $\operatorname{Spc}(\mathcal{K})$ is a spectral topological space. The homological spectrum $\operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ is only a *set*. Of course, one can pull back the topology of the former to the latter via $\pi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \twoheadrightarrow \operatorname{Spc}(\mathcal{K})$ but by [BHS23, Proposition 4.5] the resulting space $\operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ is spectral if and only if π is a bijection. Hence we prefer to state topological properties of $W^{\mathrm{h}}(\mathcal{I})$ in terms of its image $W(\mathcal{I})$ in $\operatorname{Spc}(\mathcal{K})$, particularly for properties involving spectral spaces.

3.5. *Recollection.* In a spectral space, the constructible subsets generate a topology, called the *constructible topology* or *patch topology*. In the case of $\operatorname{Spc}(\mathcal{K})$, a constructible subset is a finite union of sets of the form $C(a,b) \coloneqq \operatorname{supp}(a)^{\complement} \cap \operatorname{supp}(b) = \big\{\mathcal{P} \in \operatorname{Spc}(\mathcal{K}) \,\big|\, a \in \mathcal{P} \text{ and } b \notin \mathcal{P}\big\}$ for $a, b \in \mathcal{K}$. Since $C(a,b) \cap C(a',b') = C(a \oplus a', b \otimes b')$ the $\{C(a,b)\}_{a,b}$ form an open basis of the constructible topology and the $\{C(a,b)^{\complement}\}_{a,b}$ form a closed basis. A *proconstructible* (or *patch-closed*) subset is a closed subset for the constructible topology, *i.e.* an arbitrary intersection of subsets $C(a,b)^{\complement} = \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement} = \big\{\mathcal{P} \in \operatorname{Spc}(\mathcal{K}) \,\big|\, a \notin \mathcal{P} \text{ or } b \in \mathcal{P}\big\}$.

3.6. *Recollection.* A subset $Y \subseteq \operatorname{Spc}(\mathcal{K})$ is proconstructible and Thomason (if and) only if Y is closed with $Y^{\complement}$ quasi-compact (see [DST19, Theorem 1.5.4 (i) and Corollary 4.1.6 (ii)]).

3.7. **Proposition.** *Let $\mathcal{I} \subseteq \operatorname{mod}\text{-}\mathcal{K}$ be a Serre $\otimes$-ideal. Then $W(\mathcal{I})$ is a proconstructible subset of $\operatorname{Spc}(\mathcal{K})$; more precisely, if we let $\mathcal{M}(\mathcal{I}) \coloneqq \big\{(a,b) \in \mathcal{K} \times \mathcal{K} \,\big|\, \bar{b} \in \langle \bar{a} \rangle \text{ in } \bar{\mathcal{A}}_{\mathcal{I}}\big\}$ then*

$$W(\mathcal{I}) = \bigcap_{(a,b) \in \mathcal{M}(\mathcal{I})} \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}. \tag{3.8}$$

Proof. Let $\mathcal{H} \supseteq \mathcal{I}$ be a homological prime in $W^{\mathrm{h}}(\mathcal{I})$ and let $(a,b) \in \mathcal{M}(\mathcal{I})$, so that $\bar{b}$ belongs to the Serre $\otimes$-ideal $\langle \bar{a} \rangle$ generated by $\bar{a}$ in $\bar{\mathcal{A}}_{\mathcal{I}}$; a fortiori $\bar{b} \in \langle \bar{a} \rangle$ in the further quotient $\bar{\mathcal{A}}_{\mathcal{H}}$. This relation prevents $\bar{a} = 0$ and $\bar{b} \neq 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$, that is, we cannot have $a \in \mathrm{h}^{-1}(\mathcal{H}) = \pi(\mathcal{H})$ and $b \notin \mathrm{h}^{-1}(\mathcal{H}) = \pi(\mathcal{H})$; hence we have $\pi(\mathcal{H}) \in \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$. This proves $\subseteq$ in (3.8).

Conversely, let $\mathcal{P} \in \operatorname{Spc}(\mathcal{K})$ belong to the right-hand side of (3.8). Define $\mathcal{J} = \langle \mathcal{I} \cup \mathrm{h}(\mathcal{P}) \rangle$ the Serre $\otimes$-ideal of mod-$\mathcal{K}$ generated by $\mathcal{I}$ and $\mathcal{P}$. Let S be the $\otimes$-multiplicative class $\mathcal{K} \smallsetminus \mathcal{P}$. We claim that $\mathcal{J} \cap \mathrm{h}(S) = \varnothing$. *Ab absurdo*, if $b \in \mathcal{K} \smallsetminus \mathcal{P}$ satisfies $\mathrm{h}(b) \in \mathcal{J}$ then there exist $a \in \mathcal{P}$ such that $\mathrm{h}(b) \in \langle \mathcal{I} \cup \{\mathrm{h}(a)\} \rangle$. Modulo $\mathcal{I}$ we get $\bar{b} \in \langle \bar{a} \rangle$ in $\bar{\mathcal{A}}_{\mathcal{I}}$, that is, $(a,b) \in \mathcal{M}(\mathcal{I})$. By assumption $\mathcal{P} \in \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$ which gives us $a \notin \mathcal{P}$ or $b \in \mathcal{P}$, either of which is a contradiction. So $\mathcal{J} \cap \mathrm{h}(S) = \varnothing$ and by [Bal20b, Lemma 3.8], there exists $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ such that $\mathcal{J} \subseteq \mathcal{H}$ and $\mathcal{H} \cap \mathrm{h}(S) = \varnothing$. By construction $\mathcal{I} \subseteq \mathcal{J} \subseteq \mathcal{H}$, so $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$ and its image $\pi(\mathcal{H}) = \mathrm{h}^{-1}(\mathcal{H})$ in $\operatorname{Spc}(\mathcal{K})$ contains $\mathcal{P}$ and avoids $S = \mathcal{K} \smallsetminus \mathcal{P}$, hence equals $\mathcal{P}$. □

3.9. **Examples.** Let us discuss subsets of $\operatorname{Spc}(\mathcal{K})$ that cannot be the spectrum of a layer, because they are not proconstructible. We use Recollection 3.6. Suppose that $\operatorname{Spc}(\mathcal{K})$ has a generic point η such that $\{\eta\}$ is not open; for instance $\mathcal{K}$ could be SH^{c} (see [Bal10, Corollary 9.5(c)]), or $\mathrm{D}^{\mathrm{perf}}(R)$ for R a domain. Then $\operatorname{Spc}(\mathcal{K}) \smallsetminus \{\eta\}$ is not the spectrum of any layer.

3.10. *Remark.* We want to prove the converse to Proposition 3.7: Every proconstructible is the spectrum $W(\mathcal{I})$ of a layer. In general, there are many such $\mathcal{I}$ and we describe the *largest* one in Proposition 3.21. But we want to use the same quantifiers as for a proconstructible subset: If we choose $\mathcal{N} \subseteq \mathcal{K} \times \mathcal{K}$ then $\bigcap_{(a,b)\in\mathcal{N}} \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$ is proconstructible by Recollection 3.5. We want an $\mathcal{I}$ for every such choice of $\mathcal{N}$. We need a preparation.

3.11. **Lemma.** *For any $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ and $a \in \mathcal{K}$, exactly one of the following is true:*

(i) *The object* $\mathrm{h}(a)$ *belongs to* $\mathcal{H}$.
(ii) *The kernel* $\ker(\mathrm{h}(\mathrm{coev}_a \colon \mathbb{1} \to a^{\vee} \otimes a))$ *of coevaluation belongs to* $\mathcal{H}$.

Proof. If $\bar{a} = 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$ then (i). If $\bar{a} \neq 0$ then by Recollection 2.3 the kernel of $\mathrm{coev}_{\bar{a}} \colon \bar{\mathbb{1}} \to \bar{a}^{\vee} \otimes \bar{a}$ in $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{H}}$ belongs to the Serre $\otimes$-ideal $\operatorname{Ker}(\bar{a} \otimes -) \subsetneq \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{H}}$ which is zero; this gives (ii). □

3.12. **Lemma.** *Let $a, b \in \mathcal{K}$ and $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. Then* $\ker(\mathrm{h}(\mathrm{coev}_a)) \otimes \mathrm{h}(b)$ *belongs to $\mathcal{H}$ if and only if $\pi(\mathcal{H}) \in \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$.*

Proof. Since $\mathrm{h}(b)$ is $\otimes$-flat and $\mathcal{H}$ is a maximal Serre $\otimes$-ideal we have $\ker(\mathrm{h}(\mathrm{coev}_a)) \otimes \mathrm{h}(b) \in \mathcal{H}$ if and only if $\ker(\mathrm{h}(\mathrm{coev}_a)) \in \mathcal{H}$ or $\mathrm{h}(b) \in \mathcal{H}$. By Lemma 3.11 this is equivalent to $\mathrm{h}(a) \notin \mathcal{H}$ or $\mathrm{h}(b) \in \mathcal{H}$, which reads $\pi(\mathcal{H}) \in \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$. □

3.13. **Proposition.** *Every proconstructible subset C of* $\operatorname{Spc}(\mathcal{K})$ *is of the form $W(\mathcal{I})$ for a Serre $\otimes$-ideal $\mathcal{I}$ of* mod-$\mathcal{K}$*: For any choice of $\mathcal{N} \subseteq \mathcal{K} \times \mathcal{K}$ such that $C = \bigcap_{(a,b)\in\mathcal{N}} \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$, define a Serre $\otimes$-ideal of* mod-$\mathcal{K}$ *by the following generators*

$$\mathcal{I} = \mathcal{I}(\mathcal{N}) \coloneqq \big\langle \big\{ \ker(\mathrm{h}(\mathrm{coev}_a)) \otimes \mathrm{h}(b) \,\big|\, (a,b) \in \mathcal{N} \big\} \big\rangle.$$

Then we have $W^{\mathrm{h}}(\mathcal{I}) = \pi^{-1}(C)$ and $W(\mathcal{I}) = C$.

Proof. Let $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. We have $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$, *i.e.* $\mathcal{I} \subseteq \mathcal{H}$ if and only if $\ker(\mathrm{h}(\mathrm{coev}_a)) \otimes \mathrm{h}(b) \in \mathcal{H}$ for every $(a,b) \in \mathcal{N}$. By Lemma 3.12, this is equivalent to $\pi(\mathcal{H}) \in \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}$ for every $(a,b) \in \mathcal{N}$, or in other words $\pi(\mathcal{H}) \in \bigcap_{(a,b)\in\mathcal{N}} \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement} = C$. So we have proved $W^{\mathrm{h}}(\mathcal{I}) = \pi^{-1}(C)$, which implies $W(\mathcal{I}) = \pi(\pi^{-1}(C)) = C$ by surjectivity of π. □

3.14. *Remark.* In Proposition 3.13, one can take for $\mathcal{N}$ the largest one, namely $\mathcal{N} = \mathcal{N}(C) = \big\{(a,b) \in \mathcal{K} \times \mathcal{K} \,\big|\, C \subseteq \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\complement}\big\}$. See Recollection 3.5.

Let us discuss the special case of closed subsets (that are in particular proconstructible).

3.15. **Definition.** Let $S \subseteq \mathcal{K}$ be a $\otimes$-multiplicative class of objects in $\mathcal{K}$. The set $Z(S) \coloneqq \bigcap_{a\in S} \operatorname{supp}(a)$ is closed in $\operatorname{Spc}(\mathcal{K})$; in fact any closed subset of $\operatorname{Spc}(\mathcal{K})$ is of this form. Using the notation of Proposition 3.13, we set $\mathcal{N}_S = \{(a, \mathbb{1}) \mid a \in S\} \subseteq \mathcal{K} \times \mathcal{K}$ and we define

$$\mathcal{I}_S \coloneqq \mathcal{I}(\mathcal{N}_S) = \big\langle \{\ker(\mathrm{h}(\mathrm{coev}_a)) \mid a \in S\} \big\rangle.$$

3.16. **Corollary.** *We have* $W(\mathcal{I}_S) = Z(S) = \cap_{a\in S} \operatorname{supp}(a)$ *and* $W^{\mathrm{h}}(\mathcal{I}_S) = \pi^{-1}(Z(S))$.

Proof. Direct from Proposition 3.13 for $\mathcal{N} = \mathcal{N}_S$ since $\operatorname{supp}(\mathbb{1})^{\mathtt{c}} = \varnothing$. □

3.17. **Proposition.** *Let S be a $\otimes$-multiplicative subset of objects in $\mathcal{K}$. Then we have*

$$\mathcal{I}_S = \big\{ M \in \text{mod-}\mathcal{K} \,\big|\, \textit{there exists } a \in S \textit{ such that } \mathrm{h}(a) \otimes M = 0 \big\}. \tag{3.18}$$

Proof. Since S is $\otimes$-multiplicative, the right-hand side of (3.18) is a Serre $\otimes$-ideal of mod-$\mathcal{K}$. Take $a \in S$. By Recollection 2.3 we have $\mathrm{h}(a)\otimes\ker(\mathrm{h}(\mathrm{coev}_a)) = 0$. This proves the inclusion $\subseteq$ in (3.18). Conversely, let $M \in \text{mod-}\mathcal{K}$ and $a \in S$ with $\mathrm{h}(a) \otimes M = 0$. Let $\iota\colon M \rightarrowtail \mathrm{h}(c)$ be a monomorphism for some $c \in \mathcal{K}$ and consider the commutative square in mod-$\mathcal{K}$

$$\begin{array}{ccc} M & \xrightarrow{\quad\iota\quad} & \mathrm{h}(c) \\ {\scriptstyle M\otimes\mathrm{h}(\mathrm{coev}_a)}\downarrow & & \downarrow{\scriptstyle \mathrm{h}(c)\otimes\mathrm{h}(\mathrm{coev}_a)} \\ M \otimes \mathrm{h}(a \otimes a^\vee) & \xrightarrow[\iota\otimes\mathrm{h}(a\otimes a^\vee)]{} & \mathrm{h}(c) \otimes \mathrm{h}(a \otimes a^\vee) \end{array}$$

In the quotient $(\text{mod-}\mathcal{K})/\big\langle \ker(\mathrm{h}(\mathrm{coev}_a))\big\rangle$, the morphism $\mathrm{h}(c)\otimes\mathrm{h}(\mathrm{coev}_a)$ becomes a monomorphism, hence so does $M\otimes\mathrm{h}(\mathrm{coev}_a)$. Thus $M\otimes\mathrm{h}(a\otimes a^\vee) = 0$ forces $M \in \big\langle \ker(\mathrm{h}(\mathrm{coev}_a))\big\rangle$. □

3.19. *Remark.* With the notation of Proposition 3.17, suppose that the closed subset $Z(S) = \cap_{a\in S} \operatorname{supp}(a)$ is Thomason. Then $Z^{\mathtt{c}} = \cup_{a\in S} \operatorname{supp}(a)^{\mathtt{c}}$ is quasi-compact, hence there exists $a_0 \in S$ such that $Z(S) = \operatorname{supp}(a_0)$. In that case we have the following modification of (3.18):

$$\mathcal{I}_S = \big\{ M \in \text{mod-}\mathcal{K} \,\big|\, \text{there exists } a \in S \text{ with } \operatorname{supp}(a) = Z(S) \text{ such that } \mathrm{h}(a) \otimes M = 0 \big\}.$$

Indeed, if $\mathrm{h}(b) \otimes M = 0$ for some $b \in S$ then $\mathrm{h}(a_0 \otimes b) \otimes M = 0$ and $\operatorname{supp}(a_0 \otimes b) = Z(S)$.

To wrap up the section, we describe the largest Serre $\otimes$-ideal with a given support. It is built in the same way as the radical of an ideal in commutative algebra.

3.20. **Definition.** For $C \subseteq \operatorname{Spc}(\mathcal{K})$ we set $\mathcal{I}_{\max}(C) = \bigcap_{\mathcal{H}\in\pi^{-1}(C)} \mathcal{H}$. It is a Serre $\otimes$-ideal.

3.21. **Proposition.** *Let $C \subseteq \operatorname{Spc}(\mathcal{K})$ be a subset and $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$ a Serre $\otimes$-ideal. We have $W^{\mathrm{h}}(\mathcal{I}) \supseteq \pi^{-1}(C)$ if and only if $\mathcal{I} \subseteq \mathcal{I}_{\max}(C)$.*

Proof. If $W^{\mathrm{h}}(\mathcal{I}) \supseteq \pi^{-1}(C)$ then $\mathcal{I} \subseteq \bigcap_{\mathcal{H}\in W^{\mathrm{h}}(\mathcal{I})} \mathcal{H} \subseteq \bigcap_{\mathcal{H}\in\pi^{-1}(C)} \mathcal{H} = \mathcal{I}_{\max}(C)$. Conversely, if $\mathcal{I} \subseteq \mathcal{I}_{\max}(C)$ then for all $\mathcal{H} \in \pi^{-1}(C)$ we have $\mathcal{H} \supseteq \mathcal{I}_{\max}(C) \supseteq \mathcal{I}$. Therefore $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$. □

As announced, $\mathcal{I}_{\max}(C)$ is another solution to Proposition 3.13.

3.22. **Proposition.** *Let $C \subseteq \operatorname{Spc}(\mathcal{K})$ be a proconstructible subset. Then $W^{\mathrm{h}}(\mathcal{I}_{\max}(C)) = \pi^{-1}(C)$ and consequently $W(\mathcal{I}_{\max}(C)) = C$.*

Proof. Write $C = \bigcap_{(a,b)\in\mathcal{N}} \operatorname{supp}(a) \cup \operatorname{supp}(b)^{\mathtt{c}}$ for some subset $\mathcal{N} \subseteq \mathcal{K} \times \mathcal{K}$. The inclusion $\pi^{-1}(C) \subseteq W^{\mathrm{h}}(\mathcal{I}_{\max}(C))$ holds by Proposition 3.21 for $\mathcal{I} = \mathcal{I}_{\max}(C)$. Conversely, recall $\mathcal{I}(\mathcal{N}) = \big\langle \big\{ \ker(\mathrm{coev}_{\mathrm{h}(a)}) \otimes \mathrm{h}(b) \,\big|\, (a, b) \in \mathcal{N} \big\} \big\rangle$ from Proposition 3.13. Since the generators of $\mathcal{I}(N)$ belong to every $\mathcal{H} \in \pi^{-1}(C)$ by Lemma 3.12, we have $\mathcal{I}(\mathcal{N}) \subseteq \mathcal{I}_{\max}(C)$. We deduce from this that $W^{\mathrm{h}}(\mathcal{I}_{\max}(C)) \subseteq W^{\mathrm{h}}(\mathcal{I}(N)) = \pi^{-1}(C)$ using Proposition 3.13. □

4. Layers over Thomason subsets

We keep $\mathcal{K}$ as in Hypothesis 2.1. In Section 3 we have seen that the spectrum $W(\mathcal{I})$ of every layer $\bar{\mathcal{A}}_{\mathcal{I}}$ is a proconstructible subset of $\operatorname{Spc}(\mathcal{K})$. We now want to study the layers that live over a Thomason subset $Y \subseteq \operatorname{Spc}(\mathcal{K})$ by using the tt-ideal $\mathcal{K}_Y = \{c \in \mathcal{K} \mid \operatorname{supp}(c) \subseteq Y\}$.

4.1. **Lemma.** *Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a Thomason subset. Let $i^*\colon \text{Mod-}\mathcal{K} \to \text{Mod-}\mathcal{K}_Y$ denote the restriction along the inclusion $i\colon \mathcal{K}_Y \hookrightarrow \mathcal{K}$. Then i^* is a Gabriel quotient with kernel*

$$\operatorname{Ker}(i^*) = \{M \in \text{Mod-}\mathcal{K} \mid \mathrm{h}(a) \otimes M = 0 \text{ for all } a \in \mathcal{K}_Y\}. \tag{4.2}$$

Thus $\operatorname{Ker}(i^*)$ is a $\otimes$-ideal, and $\mathrm{h}(a) \otimes N$ is $\operatorname{Ker}(i^*)$-local for all $a \in \mathcal{K}_Y$ and $N \in \text{Mod-}\mathcal{K}$.

Proof. Since $i\colon \mathcal{K}_Y \hookrightarrow \mathcal{K}$ is fully faithful, the induced $i_!\colon \text{Mod-}\mathcal{K}_Y \to \text{Mod-}\mathcal{K}$ is fully faithful. Hence its exact right adjoint $i^*\colon \text{Mod-}\mathcal{K} \to \text{Mod-}\mathcal{K}_Y$ is a Gabriel quotient. To prove (4.2), recall that, by [BKS19, Lemma 2.5], there is a natural isomorphism of functors $\mathcal{K}^{\mathrm{op}} \to \mathrm{Ab}$

$$\mathrm{h}(a) \otimes M \cong M(a^\vee \otimes -) \tag{4.3}$$

for any $M \in \text{Mod-}\mathcal{K}$ and $a \in \mathcal{K}$. Hence if $i^*(M) = M|_{\mathcal{K}_Y} = 0$ then for every $a \in \mathcal{K}_Y$ we have $\mathrm{h}(a) \otimes M = 0$ since $a^\vee \otimes \mathcal{K} \subseteq \mathcal{K}_Y$. Conversely, if $\mathrm{h}(a) \otimes M = M(a^\vee \otimes -) = 0$ then M vanishes on $a \otimes a^\vee \otimes a$ and therefore on a itself, which is a direct summand. Hence (4.2). It follows that $\operatorname{Ker}(i^*)$ is a $\otimes$-ideal of Mod-$\mathcal{K}$. Finally, fix $a \in \mathcal{K}_Y$ and $N \in \text{Mod-}\mathcal{K}$. Then $(- \otimes \mathrm{h}(a^\vee)) \dashv (\mathrm{h}(a) \otimes -)$ is an adjunction of exact functors on $\mathcal{A} = \text{Mod-}\mathcal{K}$. Therefore

$$\operatorname{Ext}^*_{\mathcal{A}}\big(M, \mathrm{h}(a) \otimes N\big) \cong \operatorname{Ext}^*_{\mathcal{A}}\big(M \otimes \mathrm{h}(a^\vee), N\big) = 0$$

for every $M \in \operatorname{Ker}(i^*)$. This implies that $\mathrm{h}(a) \otimes N$ is $\operatorname{Ker}(i^*)$-local. □

The localizing subcategory $\operatorname{Ker}(i^*)$ is almost never generated by finitely presented objects:

4.4. **Proposition.** *With the same notation as in Lemma 4.1, the localizing subcategory $\operatorname{Ker}(i^*)$ is of finite type if and only if Y is closed and open, in which case $\mathcal{K} \cong \mathcal{K}_Y \times \mathcal{K}_{Y^{\complement}}$.*

Proof. By Theorem 2.14, $\operatorname{Ker}(i^*)$ is of finite type if and only if $i^*(\mathbb{1})$ is finitely presented in Mod-$\mathcal{K}_Y$. So suppose first that $i^*(\mathbb{1}) \in \text{mod-}\mathcal{K}_Y$. Note that $i^*(\mathbb{1})$ remains a cohomological functor, so it is injective in mod-$\mathcal{K}_Y$ by [Kra00, Lemma 2.7]. Hence $i^*(\mathbb{1}) \simeq \mathrm{h}(e)$ for some $e \in \mathcal{K}_Y$ (Recollection 2.4). It follows (see Recollection 2.16) that $\mathbb{e}_Y = i_! i^*(\mathbb{1}) \simeq i_!(\mathrm{h}(e)) \cong \mathrm{h}(i(e))$ is finitely presented in $\mathcal{A}$, and therefore so is $\mathbb{f}_Y$ by (2.18). Since $\mathbb{f}_Y$ is also cohomological, we get $\mathbb{f}_Y \cong \mathrm{h}(f)$ for some $f \in \mathcal{K}$ by [Kra00, Lemma 2.7] again. Then $\operatorname{Hom}(\Sigma^{-1}\mathbb{f}_Y, \mathbb{e}_Y) = \operatorname{Hom}(\mathbb{f}_Y, \Sigma\mathbb{e}_Y) = 0$ by $\mathbb{e}_Y \otimes \mathbb{f}_Y = 0$ (2.19) and rigidity. Thus there is a short exact sequence $0 \to \mathrm{h}(e) \to \mathbb{1} \to \mathrm{h}(f) \to 0$. It easily follows that $\mathbb{1} \simeq e \oplus f$ in $\mathcal{K}$, forcing $\mathcal{K} \cong \mathcal{K}_Y \times \mathcal{K}_{Y^{\complement}}$ with $Y^{\complement} = \operatorname{supp}(f)$ closed. The reverse direction is clear: $i^*(\mathbb{1}) \cong \mathrm{h}(\mathbb{1}_{\mathcal{K}_Y})$. □

4.5. **Definition.** Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be Thomason. We denote by $\mathcal{B}_{\min}(Y)$ the localizing $\otimes$-ideal of Mod-$\mathcal{K}$ of Lemma 4.1 (the notation $\mathcal{B}_{\min}$ is justified by Theorem 4.6 and Corollary 4.7):

$$\mathcal{B}_{\min}(Y) \coloneqq \operatorname{Ker}(\text{Mod-}\mathcal{K} \overset{i^*}{\twoheadrightarrow} \text{Mod-}\mathcal{K}_Y) = \{M \in \text{Mod-}\mathcal{K} \mid \mathrm{h}(a) \otimes M = 0 \text{ for all } a \in \mathcal{K}_Y\}.$$

4.6. **Theorem.** *Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a Thomason subset and $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$ a Serre $\otimes$-ideal. We have $W^{\mathrm{h}}(\mathcal{I}) \subseteq \pi^{-1}(Y)$ if and only if $\mathcal{B}_{\min}(Y) \subseteq \operatorname{Loc}(\mathcal{I})$.*

Proof. First, suppose that $W^{\mathrm{h}}(\mathcal{I}) \subseteq \pi^{-1}(Y)$. For every homological prime $\mathcal{H} \in \pi^{-1}(Y)$, we have $\overline{\mathbb{f}_Y} = 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$ by Lemma 2.20 (a). So $\mathbb{f}_Y$ is zero in the residue fields $\bar{\mathcal{A}}_{\mathcal{H}}$ for all $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$. As $\mathbb{f}_Y$ is a $\otimes$-flat ring object, it must go to zero in $\bar{\mathcal{A}}_{\mathcal{I}}$ by the Nilpotence Theorem (Recollection 3.3). Therefore $\overline{\mathbb{e}_Y} \simeq \bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{I}}$ by (2.18). Let $M \in \mathcal{B}_{\min}(Y)$. By (4.2) we have

$M \otimes \mathrm{h}(a) = 0$ for every $a \in \mathcal{K}_Y$. Since $\mathbb{e}_Y$ is a colimit of $\mathrm{h}(a)$ for $a \in \mathcal{K}_Y$, we have $M \otimes \mathbb{e}_Y = 0$ as well. Therefore in $\bar{\mathcal{A}}_{\mathcal{I}}$, we have $\bar{M} \cong \bar{M} \otimes \bar{\mathbb{1}} \simeq \bar{M} \otimes \overline{\mathbb{e}_Y} = 0$, proving that $M \in \mathrm{Loc}(\mathcal{I})$.

Conversely, suppose that $W^{\mathrm{h}}(\mathcal{I}) \not\subseteq \pi^{-1}(Y)$. Then there exists $\mathcal{H} \supseteq \mathcal{I}$ with $\pi(\mathcal{H}) \notin Y$. We have $\kappa(\mathcal{H}) \notin \mathrm{Loc}(\mathcal{I})$ by Lemma 3.2. On the other hand, because $\pi(\mathcal{H}) \notin Y$, we must have $\mathrm{h}(a) \in \mathcal{H}$ for all $a \in \mathcal{K}_Y$. By the projection formula for $Q_{\mathcal{H}} \dashv R_{\mathcal{H}}$, for all $a \in \mathcal{K}_Y$, we get in $\mathcal{A}$

$$\kappa(\mathcal{H}) \otimes \mathrm{h}(a) = R_{\mathcal{H}}\left(\overline{\kappa(\mathcal{H})}\right) \otimes \mathrm{h}(a) \cong R_{\mathcal{H}}\left(\overline{\kappa(\mathcal{H})} \otimes Q_{\mathcal{H}}(\mathrm{h}(a))\right) = 0.$$

Therefore by (4.2), we have $\kappa(\mathcal{H}) \in \mathcal{B}_{\min}(Y)$. Hence we have proved that $\mathcal{B}_{\min}(Y)$ is not contained in $\mathrm{Loc}(\mathcal{I})$ since the former contains $\kappa(\mathcal{H})$ but the latter does not. □

We can combine the above with Proposition 3.21 when Y is furthermore proconstructible (see Recollection 3.6). We obtain an 'interval' of Serre $\otimes$-ideals whose spectrum is Y.

4.7. **Corollary.** *Let $Y \subseteq \mathrm{Spc}(\mathcal{K})$ be a closed subset with quasi-compact complement and let $\mathcal{I} \subseteq \mathrm{mod}\text{-}\mathcal{K}$ be a Serre $\otimes$-ideal. Consider the two localizing $\otimes$-ideals of $\mathcal{A} = \mathrm{Mod}\text{-}\mathcal{K}$*

$$\mathcal{B}_{\min}(Y) = \mathrm{Ker}(i^*\colon \mathrm{Mod}\text{-}\mathcal{K} \twoheadrightarrow \mathrm{Mod}\text{-}\mathcal{K}_Y) \qquad \text{and} \qquad \mathcal{B}_{\max}(Y) \coloneqq \mathrm{Loc}(\mathcal{I}_{\max}(Y)).$$

Then we have $W^{\mathrm{h}}(\mathcal{I}) = \pi^{-1}(Y)$ if and only if $\mathcal{B}_{\min}(Y) \subseteq \mathrm{Loc}(\mathcal{I}) \subseteq \mathcal{B}_{\max}(Y)$.

Proof. By Proposition 3.21, $W^{\mathrm{h}}(\mathcal{I}) \supseteq \pi^{-1}(Y)$ if and only if $\mathcal{I} \subseteq \mathcal{I}_{\max}(Y)$, that is, $\mathrm{Loc}(\mathcal{I}) \subseteq \mathcal{B}_{\max}(Y)$. By Theorem 4.6, $W^{\mathrm{h}}(\mathcal{I}) \subseteq \pi^{-1}(Y)$ if and only if $\mathcal{B}_{\min}(Y) \subseteq \mathrm{Loc}(\mathcal{I})$. □

4.8. *Remark.* Although $\mathcal{B}_{\max}(Y)$ is of finite type, the subcategory $\mathcal{B}_{\min}(Y)$ need not be (Proposition 4.4). This may explain why $\mathcal{B}_{\min}(Y)$ does not have an analog in commutative algebra.

5. Preservation of layers

For the whole section, $F\colon \mathcal{K} \to \mathcal{L}$ is a tt-functor between two rigid idempotent-complete tt-categories $\mathcal{K}$ and $\mathcal{L}$, as in Hypothesis 2.1. We denote the induced map on spectra by

$$\varphi = \mathrm{Spc}(F)\colon \mathrm{Spc}(\mathcal{L}) \to \mathrm{Spc}(\mathcal{K}).$$

Recall that $\mathrm{supp}_{\mathcal{L}}(F(c)) = \varphi^{-1}(\mathrm{supp}_{\mathcal{K}}(c))$ for all $c \in \mathcal{K}$. It follows that the preimage $\varphi^{-1}(Y)$ of a Thomason subset Y is Thomason and that $F(\mathcal{K}_Y) \subseteq \mathcal{L}_{\varphi^{-1}(Y)}$.

5.1. **Definition.** Let $Y \subseteq \mathrm{Spc}(\mathcal{K})$ be a Thomason subset. We say that F is an *equivalence over Y* if the restriction of F on $\mathcal{K}_Y$ is an equivalence

$$F_Y \coloneqq F|_{\mathcal{K}_Y}\colon \mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{\varphi^{-1}(Y)}.$$

(One might prefer to call such an F an 'equivalence *with supports in Y*' since $\mathcal{K}_Y$ should not be mistaken for an honest tt-category. In particular, by Remark 2.15, if $Y \hookrightarrow X$ is a closed subscheme and $\mathcal{K} = \mathrm{D}^{\mathrm{perf}}(X)$ then $\mathcal{K}_Y$ should not be confused with $\mathrm{D}^{\mathrm{perf}}(Y)$.)

The following result also holds for $\mathrm{Spc}^{\mathrm{h}}$; see Corollary 5.14.

5.2. **Proposition.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over Y. Then φ restricts to a continuous bijection $\varphi^{-1}(Y) \xrightarrow{\sim} Y$. Moreover, if Z is a Thomason subset of $\mathrm{Spc}(\mathcal{L})$ contained in $\varphi^{-1}(Y)$ then $\varphi(Z)$ is a Thomason subset of $\mathrm{Spc}(\mathcal{K})$. Finally, if Y is closed with quasi-compact complement then this bijection $\varphi^{-1}(Y) \xrightarrow{\sim} Y$ is a homeomorphism.*

Proof. Write $Y' = \varphi^{-1}(Y)$ and consider $\varphi|_{Y'} \colon Y' \to Y$. Since F detects nilpotence on $\mathcal{K}_Y$ it follows from [Bal18] that $\varphi|_{Y'}$ is surjective. For injectivity, we adapt [BS25, Lemma 4.5]. Let $\mathcal{Q}, \mathcal{Q}' \in Y'$ satisfy $\varphi(\mathcal{Q}) = \varphi(\mathcal{Q}') \in Y$. Let $c \in \mathcal{K}_Y$ such that $\varphi(\mathcal{Q}) = \varphi(\mathcal{Q}') \in \operatorname{supp}(c)$. This implies that $F(c) \notin \mathcal{Q}'$. Let $d \in \mathcal{Q}$. Then we have $d \otimes F(c) \in \mathcal{L}_{Y'}$. By our equivalence $\mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{Y'}$, there exists $b \in \mathcal{K}_Y$ such that $F(b) \simeq d \otimes F(c) \in \mathcal{Q}$. Hence $b \in F^{-1}(\mathcal{Q}) = F^{-1}(\mathcal{Q}')$ and therefore $d \otimes F(c) \simeq F(b) \in \mathcal{Q}'$. Since $F(c) \notin \mathcal{Q}'$ we have $d \in \mathcal{Q}'$. In short $\mathcal{Q} \subseteq \mathcal{Q}'$. By symmetry, $\mathcal{Q} = \mathcal{Q}'$. So $\varphi|_{Y'}$ is a continuous bijection.

Let $d \in \mathcal{L}_{Y'}$ and $c \in \mathcal{K}_Y$ such that $F(c) \simeq d$. Then $\varphi^{-1}(\operatorname{supp}(c)) = \operatorname{supp}(d)$ implies that $\operatorname{supp}(c) = \varphi(\operatorname{supp}(d))$. A Thomason subset $Z \subseteq \operatorname{Spc}(\mathcal{L})$ contained in Y' is a union of $\operatorname{supp}(d)$ with $d \in \mathcal{L}_{Y'}$. This gives the 'moreover part'. When Y, hence Y', is closed Thomason (*i.e.* the support of an object), then these $\{\operatorname{supp}(d)\}_{d \in \mathcal{L}_{Y'}}$ form a basis of the subspace topology on Y'. Hence the final statement. □

We want to show that equivalences over Thomason subsets localize well.

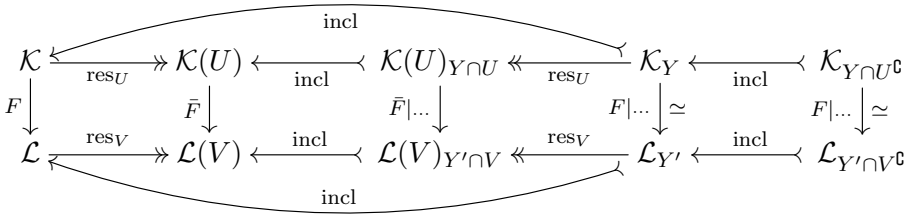
5.3. *Recollection.* If $U \subseteq \operatorname{Spc}(\mathcal{K})$ is the complement of a Thomason subset (*i.e.* an intersection of quasi-compact open subsets), then $\mathcal{K}(U)$ denotes $(\mathcal{K}/\mathcal{K}_{U^\complement})^\natural$ with the obvious functor $\operatorname{res}_U \colon \mathcal{K} \twoheadrightarrow \mathcal{K}(U)$. One has $\operatorname{Spc}(\mathcal{K}(U)) \cong U$ and $\operatorname{supp}(\operatorname{res}_U(c)) = \operatorname{supp}(c) \cap U$ for all $c \in \mathcal{K}$.

5.4. **Proposition.** *Let $F \colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over a Thomason Y. Let $U \subseteq \operatorname{Spc}(\mathcal{K})$ and $V \subseteq \operatorname{Spc}(\mathcal{L})$ with $U^\complement$ and $V^\complement$ Thomason. Suppose that $V \subseteq \varphi^{-1}(U)$ so that F restricts to $\bar{F} \colon \mathcal{K}(U) \to \mathcal{L}(V)$. If $\bar{F}$ is conservative then it is an equivalence over $Y \cap U$.*

Proof. Since we assume $\varphi^{-1}(U^\complement) \subseteq V^\complement$ we have $F(\mathcal{K}_{U^\complement}) \subseteq \mathcal{L}_{V^\complement}$ and F descends to a tt-functor $\bar{F} \colon \mathcal{K}(U) \to \mathcal{L}(V)$ such that $\operatorname{res}_V \circ F \cong \bar{F} \circ \operatorname{res}_U$. Let $Y' = \varphi^{-1}(Y)$. We have

$$\mathcal{K}(U)_{Y \cap U} \cong (\mathcal{K}_Y/\mathcal{K}_{Y \cap U^\complement})^\natural \qquad \text{and} \qquad \mathcal{L}(V)_{Y' \cap V} \cong (\mathcal{L}_{Y'}/\mathcal{L}_{Y' \cap V^\complement})^\natural. \tag{5.5}$$

Indeed, if $Z := U^\complement$ then $\mathcal{K}_{Y \cap Z} = \mathcal{K}_Y \cap \mathcal{K}_Z$ and the functor $(\mathcal{K}_Y/\mathcal{K}_{Y \cap Z})^\natural \to (\mathcal{K}/\mathcal{K}_Z)^\natural = \mathcal{K}(U)$ induced by res_U is fully faithful by [Ver96, Chapitre II, Proposition 2.3.5]. Its image is a tt-ideal determined by its support, which is obviously contained in $Y \cap U$ since $\operatorname{supp}(\operatorname{res}_U(c)) = \operatorname{supp}(c) \cap U$. Conversely, for every $d \in \mathcal{K}(U)$ there exists $c \in \mathcal{K}$ such that $\operatorname{res}_U(c) \simeq d \oplus \Sigma(d)$; if $d \in \mathcal{K}(U)_{Y \cap U}$ we get $\operatorname{supp}(c) \subseteq \operatorname{supp}(d) \cup Z \subseteq Y \cup Z$. Since $\mathcal{K}_{Y \cup Z} = \mathcal{K}_Y \vee \mathcal{K}_Z$ has the same image in $\mathcal{K}(U)$ as $\mathcal{K}_Y$, we get the first equivalence in (5.5). The second is the same in $\mathcal{L}$. We claim that we have a commutative diagram as follows (where $|_{\dots}$ indicates restriction)



The functor $\bar{F}$ sends $\mathcal{K}(U)_{Y \cap U}$ into $\mathcal{L}(V)_{Y' \cap V}$ thanks to (5.5) and the fact that for every $c \in \mathcal{K}_Y$, we have $\operatorname{supp}_{\mathcal{L}(V)}(\bar{F}(\operatorname{res}_U(c))) = \operatorname{supp}_{\mathcal{L}(V)}(\operatorname{res}_V(F(c))) = \varphi^{-1}(\operatorname{supp}(c)) \cap V \subseteq Y' \cap V$. In the right-most square, the equivalence $F_Y := F|_{\mathcal{K}_Y} \colon \mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{Y'}$ sends the subcategory $\mathcal{K}_{Y \cap U^\complement}$ exactly onto $\mathcal{L}_{Y' \cap V^\complement}$ for these subcategories are respectively $\operatorname{Ker}(\operatorname{res}_U) \cap \mathcal{K}_Y$ and $\operatorname{Ker}(\operatorname{res}_V) \cap \mathcal{L}_{Y'}$ by (5.5) again, and $0 = \operatorname{res}_V(F_Y(c)) = \bar{F}(\operatorname{res}_U(c))$ if and only if $\operatorname{res}_U(c) = 0$ by our conservativity hypothesis on $\bar{F}$. It follows from (5.5) that F_Y descends to an equivalence $\mathcal{K}(U)_{Y \cap U} \xrightarrow{\sim} \mathcal{L}(V)_{Y' \cap V}$, which is necessarily $\bar{F}|_{\mathcal{K}(U)_{Y \cap U}}$ by the above diagram. □

5.6. **Lemma.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over Y. Then F yields a natural isomorphism $\alpha_{b,c}\colon \operatorname{Hom}_{\mathcal{K}}(b,c) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{L}}(F(b),F(c))$ for every $b \in \mathcal{K}_Y$ and $c \in \mathcal{K}$.*

Proof. Naturality is easy. Then, since every $b \in \mathcal{K}$ is a retract of $b \otimes b^\vee \otimes b$, it suffices to prove that $\alpha_{b\otimes b^\vee \otimes b\,,\,c}$ is an isomorphism. One verifies that $\alpha_{b\otimes b^\vee\otimes b,c}$ factors as $\operatorname{Hom}_{\mathcal{K}}(b\otimes b^\vee\otimes b, c) \cong \operatorname{Hom}_{\mathcal{K}}(b\otimes b, b\otimes c) \xrightarrow[\alpha_{b\otimes b,b\otimes c}]{\sim} \operatorname{Hom}_{\mathcal{L}}(F(b\otimes b), F(b\otimes c)) \cong \operatorname{Hom}_{\mathcal{L}}(F(b\otimes b^\vee\otimes b), F(c))$ where the unlabelled isomorphisms are given by rigidity and $\alpha_{b\otimes b,b\otimes c}$ is an isomorphism since $b\otimes c \in \mathcal{K}_Y$ and $F|_{\mathcal{K}_Y}$ is fully faithful by assumption. Hence the result. □

5.7. **Lemma.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over Y. Let $Y' = \varphi^{-1}(Y) \subseteq \operatorname{Spc}(\mathcal{L})$ be the preimage of Y. Then the following diagram commutes up to isomorphism:*

$$\begin{array}{ccc} \text{Mod-}\mathcal{K} & \xrightarrow{F_!} & \text{Mod-}\mathcal{L} \\ {\scriptstyle i^*}\downarrow & & \downarrow{\scriptstyle j^*} \\ \text{Mod-}\mathcal{K}_Y & \xrightarrow[\simeq]{(F_Y)_!} & \text{Mod-}\mathcal{L}_{Y'} \end{array} \tag{5.8}$$

where $i\colon \mathcal{K}_Y \hookrightarrow \mathcal{K}$ and $j\colon \mathcal{L}_{Y'} \hookrightarrow \mathcal{L}$ are the inclusions and where $F_Y = F|_{\mathcal{K}_Y}\colon \mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{Y'}$.

Proof. This is not trivial since we mix $(-)^*$ and $(-)_!$. As $F_Y\colon \mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{Y'}$ is an equivalence, so is $(F_Y)_!$, and its inverse is its adjoint $(F_Y)^*$. Hence it suffices to construct an isomorphism

$$i^* \xrightarrow{\sim} (F_Y)^* \circ j^* \circ F_!\colon \text{Mod-}\mathcal{K} \to \text{Mod-}\mathcal{K}_Y.$$

All four functors are cocontinuous and exact. By the universal property of Mod-$\mathcal{K}$ ([Kra00, Lemma 2.2]), it suffices to give such an isomorphism after precomposing with the Yoneda embedding $\mathrm{h}\colon \mathcal{K} \hookrightarrow \text{Mod-}\mathcal{K}$. For every $c \in \mathcal{K}$, we have a natural transformation α in Mod-$\mathcal{K}_Y$

$$\alpha_{-,c}\colon i^*(\mathrm{h}(c)) = \operatorname{Hom}_{\mathcal{K}}(-,c)|_{\mathcal{K}_Y} \xrightarrow{F} \operatorname{Hom}_{\mathcal{L}}(F(-),F(c))|_{\mathcal{K}_Y} = (F_Y)^*(j^*(F_!(\mathrm{h}(c)))).$$

So it suffices to know that for all $c \in \mathcal{K}$ and all $b \in \mathcal{K}_Y$ the functor F yields a bijection $\alpha_{b,c}\colon \operatorname{Hom}_{\mathcal{K}}(b,c) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{L}}(F(b),F(c))$, which is what we proved in Lemma 5.6. □

Recall $W^{\mathrm{h}}(\mathcal{I}) = \{\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \mid \mathcal{I} \subseteq \mathcal{H}\}$ and $W(\mathcal{I}) = \pi(W^{\mathrm{h}}(\mathcal{I}))$ from Definition 3.1 and recall that $\pi\colon \operatorname{Spc}^{\mathrm{h}} \twoheadrightarrow \operatorname{Spc}$ is often a bijection in examples (Recollection 2.7). Therefore, in first approximation, the reader can ignore π below and think $W^{\mathrm{h}}(\mathcal{I}) = W(\mathcal{I})$.

5.9. **Theorem.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor and $\varphi = \operatorname{Spc}(F)$. Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a Thomason subset such that F is an equivalence over Y (Definition 5.1). Let $Y' = \varphi^{-1}(Y)$. Then, using $F_!\colon \text{mod-}\mathcal{K} \to \text{mod-}\mathcal{L}$, we get an inclusion-preserving bijection*

$$\left\{ \begin{array}{c} \text{Serre } \otimes\text{-ideals } \mathcal{I} \text{ of mod-}\mathcal{K} \\ \text{such that } W^{\mathrm{h}}(\mathcal{I}) \subseteq \pi^{-1}(Y) \end{array} \right\} \underset{\simeq}{\overset{\Psi}{\longleftarrow}} \left\{ \begin{array}{c} \text{Serre } \otimes\text{-ideals } \mathcal{J} \text{ of mod-}\mathcal{L} \\ \text{such that } W^{\mathrm{h}}(\mathcal{J}) \subseteq \pi^{-1}(Y') \end{array} \right\}$$

given by $\Psi(\mathcal{J}) = (F_!)^{-1}(\mathcal{J})$. Moreover, if $\mathcal{I} = \Psi(\mathcal{J})$ then F induces an equivalence

$$\frac{\text{Mod-}\mathcal{K}}{\operatorname{Loc}(\mathcal{I})} \xrightarrow{\sim} \frac{\text{Mod-}\mathcal{L}}{\operatorname{Loc}(\mathcal{J})} \tag{5.10}$$

between the $\mathcal{I}$-layer of $\mathcal{K}$ and the $\mathcal{J}$-layer of $\mathcal{L}$.

Proof. The preimage of a localizing $\otimes$-ideal under a cocontinuous exact tensor-functor of tensor Grothendieck categories is again a localizing $\otimes$-ideal. Using $F_!\colon \mathrm{Mod}\text{-}\mathcal{K} \to \mathrm{Mod}\text{-}\mathcal{L}$ and $j^* \circ F_! \cong (F_Y)_! \circ i^*$ from Lemma 5.7, we get a commutative diagram (of posets)

$$\begin{array}{ccc} \left\{\begin{array}{c}\text{Localizing } \otimes\text{-ideals of Mod-}\mathcal{K}\\ \text{containing } \mathcal{B}_{\min}(Y) = \mathrm{Ker}(i^*)\end{array}\right\} & \xleftarrow[=:\Psi]{(F_!)^{-1}(-)} & \left\{\begin{array}{c}\text{Localizing } \otimes\text{-ideals of Mod-}\mathcal{L}\\ \text{containing } \mathcal{B}_{\min}(Y') = \mathrm{Ker}(j^*)\end{array}\right\} \\ {\scriptstyle (i^*)^{-1}(-)}\uparrow{\scriptstyle \simeq} & & {\scriptstyle \simeq}\uparrow{\scriptstyle (j^*)^{-1}(-)} \\ \{\text{ Localizing } \otimes\text{-ideals of Mod-}\mathcal{K}_Y\ \} & \xleftarrow[\simeq]{((F_Y)_!)^{-1}(-)} & \{\text{ Localizing } \otimes\text{-ideals of Mod-}\mathcal{L}_{Y'}\ \} \end{array} \tag{5.11}$$

involving the big categories of (not necessarily finitely presented) modules. The bottom map is a bijection since $F_Y = F|_{\mathcal{K}_Y}\colon \mathcal{K}_Y \xrightarrow{\sim} \mathcal{L}_{Y'}$ is an equivalence. The vertical maps are bijections since $\mathrm{Mod}\text{-}\mathcal{K}_Y = (\mathrm{Mod}\text{-}\mathcal{K})/\mathcal{B}_{\min}(Y)$ by Lemma 4.1 and similarly on the $\mathcal{L}$-side. Let us denote by Ψ the resulting bijection $\Psi(\mathcal{C}) = (F_!)^{-1}(\mathcal{C})$ at the top of (5.11). (We shall show that it 'restricts' to the Ψ of the statement.) Whenever $\mathcal{B}$ corresponds to $\mathcal{C}$ under this bijection, the functor $F_!\colon \mathrm{Mod}\text{-}\mathcal{K} \to \mathrm{Mod}\text{-}\mathcal{L}$ descends to an equivalence

$$\frac{\mathrm{Mod}\text{-}\mathcal{K}}{\mathcal{B}} \xrightarrow{\ \sim\ } \frac{\mathrm{Mod}\text{-}\mathcal{L}}{\mathcal{C}}. \tag{5.12}$$

Indeed, by the above discussion (and Lemma 5.7 again) this functor is simply the equivalence $(F_Y)_!$ descended to the corresponding quotients of $\mathrm{Mod}\text{-}\mathcal{K}_Y$ and $\mathrm{Mod}\text{-}\mathcal{L}_{Y'}$. Since this is a $\otimes$-functor (as $F_!$ is) the $\otimes$-unit is finitely presented on one side of (5.12) if and only if it is on the other. In other words, the Ψ of (5.11) restricts to a bijection on the *finite type* localizing $\otimes$-ideals by Theorem 2.14. This gives the bijection at the top of the following diagram:

$$\begin{array}{ccc} \left\{\begin{array}{c}\text{Localizing } \otimes\text{-ideals of Mod-}\mathcal{K}\\ \text{of \textit{finite type} containing } \mathcal{B}_{\min}(Y)\end{array}\right\} & \xleftarrow[\simeq]{\Psi} & \left\{\begin{array}{c}\text{Localizing } \otimes\text{-ideals of Mod-}\mathcal{L}\\ \text{of \textit{finite type} containing } \mathcal{B}_{\min}(Y')\end{array}\right\} \\ {\scriptstyle \simeq}\downarrow{\scriptstyle (-)^{\mathrm{fp}}=(-)\cap \mathrm{mod}\text{-}\mathcal{K}} & & {\scriptstyle \simeq}\downarrow{\scriptstyle (-)^{\mathrm{fp}}=(-)\cap \mathrm{mod}\text{-}\mathcal{L}} \\ \left\{\begin{array}{c}\text{Serre } \otimes\text{-ideals } \mathcal{I} \text{ of mod-}\mathcal{K}\\ \text{such that } W^{\mathrm{h}}(\mathcal{I}) \subseteq \pi^{-1}(Y)\end{array}\right\} & \xleftarrow{\ \cdots\ } & \left\{\begin{array}{c}\text{Serre } \otimes\text{-ideals } \mathcal{J} \text{ of mod-}\mathcal{L}\\ \text{such that } W^{\mathrm{h}}(\mathcal{J}) \subseteq \pi^{-1}(Y')\end{array}\right\} \end{array}$$

The vertical maps on both sides are well-defined bijections by Theorem 4.6 and what it means to be of finite type. (Their inverses are given by $\mathrm{Loc}(-)$.) So we get the (dotted) bijection at the bottom and we claim it is the Ψ of the statement. Indeed for every localizing $\otimes$-ideal $\mathcal{C}$ of $\mathrm{Mod}\text{-}\mathcal{L}$, which is of finite type and contains $\mathcal{B}_{\min}(Y')$, the following Serre $\otimes$-ideals of $\mathrm{mod}\text{-}\mathcal{K}$ coincide because $F_!$ preserves finitely presented objects:

$$\Psi(\mathcal{C}) \cap \mathrm{mod}\text{-}\mathcal{K} = \big((F_!)^{-1}(\mathcal{C})\big) \cap \mathrm{mod}\text{-}\mathcal{K} = \big((F_!)^{-1}(\mathcal{C} \cap \mathrm{mod}\text{-}\mathcal{L})\big) \cap \mathrm{mod}\text{-}\mathcal{K}.$$

This gives the bijection Ψ announced in the statement (where we only used $F_!$ on finitely presented modules). Finally, (5.10) follows from (5.12) for $\mathcal{B} = \mathrm{Loc}(\mathcal{I})$ and $\mathcal{C} = \mathrm{Loc}(\mathcal{J})$. □

5.13. *Remark.* One can trace through the above proof to show that Ψ^{-1} maps a Serre $\otimes$-ideal $\mathcal{I}$ to the finitely presented part of the localizing $\otimes$-ideal generated by $F_!(\mathcal{I}) \cup \mathcal{B}_{\min}(Y')$.

Let us denote by $\varphi^{\mathrm{h}} = \mathrm{Spc}^{\mathrm{h}}(F)\colon \mathrm{Spc}^{\mathrm{h}}(\mathcal{L}) \to \mathrm{Spc}^{\mathrm{h}}(\mathcal{K})$ the map induced by F on homological spectra, given by preimages $\varphi^{\mathrm{h}}(\mathcal{G}) = (F_!)^{-1}(\mathcal{G})$ under $F_!\colon \mathrm{mod}\text{-}\mathcal{K} \to \mathrm{mod}\text{-}\mathcal{L}$. Even in the absence of the steel property, the bijection $\varphi^{-1}(Y) \xrightarrow{\sim} Y$ of Proposition 5.2 extends to $\mathrm{Spc}^{\mathrm{h}}$.

5.14. **Corollary.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over a Thomason subset $Y \subseteq \operatorname{Spc}(\mathcal{K})$ and let $Y' \subseteq \operatorname{Spc}(\mathcal{L})$ be its preimage. Then $\varphi^{\mathrm{h}}\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{L}) \to \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ restricts to a bijection $\pi^{-1}(Y') \xrightarrow{\sim} \pi^{-1}(Y)$. Moreover, for every $\mathcal{G} \in \pi^{-1}(Y')$ the functor F induces an equivalence between the homological residue fields at $\mathcal{G}$ and at $\mathcal{H} = \varphi^{\mathrm{h}}(\mathcal{G})$*

$$\frac{\text{Mod-}\mathcal{K}}{\operatorname{Loc}(\mathcal{H})} \xrightarrow{\sim} \frac{\text{Mod-}\mathcal{L}}{\operatorname{Loc}(\mathcal{G})}.$$

Proof. Apply Theorem 5.9. When $\mathcal{I} = \Psi(\mathcal{J})$, the equivalence (5.10) on the quotients shows that $\mathcal{I}$ is maximal (a homological prime) if and only if $\mathcal{J}$ is maximal. For $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ we have $W^{\mathrm{h}}(\mathcal{H}) = \{\mathcal{H}\}$ and so the condition $W^{\mathrm{h}}(\mathcal{H}) \subseteq \pi^{-1}(Y)$ simply reads $\mathcal{H} \in \pi^{-1}(Y)$. Similarly, $W^{\mathrm{h}}(\mathcal{G}) \subseteq \pi^{-1}(Y')$ if and only if $\mathcal{G} \in \pi^{-1}(Y')$. Since $\Psi(\mathcal{G}) = (F_!)^{-1}(\mathcal{G}) = \varphi^{\mathrm{h}}(\mathcal{G})$, we see that the bijection Ψ restricts to the claimed bijection $\varphi^{\mathrm{h}}\colon \pi^{-1}(Y') \xrightarrow{\sim} \pi^{-1}(Y)$. □

5.15. *Remark.* Since the bijection Ψ of Theorem 5.9 preserves inclusions and restricts to φ^{h} on maximal ideals, we see that whenever $\mathcal{I} = \Psi(\mathcal{J})$ then $\varphi^{\mathrm{h}}(W^{\mathrm{h}}(\mathcal{J})) = W^{\mathrm{h}}(\mathcal{I})$, and thus

$$(\varphi^{\mathrm{h}})^{-1}(W^{\mathrm{h}}(\mathcal{I})) = W^{\mathrm{h}}(\mathcal{J}). \tag{5.16}$$

5.17. *Remark.* The equality (5.16) gives a specialization of Theorem 5.9 when Y (hence Y') is proconstructible. Since $(\varphi^{\mathrm{h}})^{-1}(\pi^{-1}(Y)) = \pi^{-1}(Y')$, the bijection Ψ restricts to a bijection between those $\mathcal{I}$ with $W^{\mathrm{h}}(\mathcal{I}) = \pi^{-1}(Y)$ and those $\mathcal{J}$ with $W^{\mathrm{h}}(\mathcal{J}) = \pi^{-1}(Y')$.

Here is another formulation of the above preservation of homological residue fields, that we shall need in Section 6. We keep notation as above.

5.18. **Corollary.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor that is an equivalence over Y. Set $Y' = \varphi^{-1}(Y)$. Let $\mathcal{J} \subseteq \text{mod-}\mathcal{L}$ be a Serre $\otimes$-ideal such that $W^{\mathrm{h}}(\mathcal{J}) \subseteq \pi^{-1}(Y')$ and $\mathcal{I} = (F_!)^{-1}(\mathcal{J})$ its correspondent in* mod-$\mathcal{K}$. *Then we have an isomorphism $F^*(E_{\mathcal{J}}) \simeq E_{\mathcal{I}}$ in* Mod-$\mathcal{K}$. *In particular, if $\mathcal{G} \in \pi^{-1}(Y')$ in $\operatorname{Spc}^{\mathrm{h}}(\mathcal{L})$ and $\mathcal{H} \coloneqq \varphi^{\mathrm{h}}(\mathcal{G})$ then $F^*(\kappa(\mathcal{G})) \simeq \kappa(\mathcal{H})$.*

Proof. By Theorem 5.9, we have an equivalence of layers, fitting in the commutative diagram

$$\begin{array}{ccc} \text{Mod-}\mathcal{K} & \xrightarrow{F_!} & \text{Mod-}\mathcal{L} \\ {\scriptstyle Q_{\mathcal{I}}}\downarrow & & \downarrow{\scriptstyle Q_{\mathcal{J}}} \\ \text{Mod-}\mathcal{K}/\operatorname{Loc}(\mathcal{I}) & \xrightarrow[\simeq]{\overline{F_!}} & \text{Mod-}\mathcal{L}/\operatorname{Loc}(\mathcal{J}). \end{array}$$

On right adjoints, we get $F^* \circ R_{\mathcal{J}} \cong R_{\mathcal{I}} \circ (\overline{F_!})^{-1}$. Applying this isomorphism to the object $\bar{E}_{\mathcal{J}}$ we get $F^*(E_{\mathcal{J}}) \simeq E_{\mathcal{I}}$ since the equivalence $\overline{F_!}$ identifies the injective hulls $\bar{E}_{\mathcal{I}}$ and $\bar{E}_{\mathcal{J}}$. □

5.19. *Remark.* The bijection Ψ of Theorem 5.9 preserves the Serre $\otimes$-ideals of the form $\mathcal{I}_S$ as in Definition 3.15. Indeed, let $S \subseteq \mathcal{K}$ be a $\otimes$-multiplicative class of objects in $\mathcal{K}$. By Corollary 3.16 the homological spectrum of $\mathcal{I}_S$ is $\pi^{-1}(Z(S))$ where $Z(S) = \cap_{a \in S} \operatorname{supp}(a)$ is the closed subset of $\operatorname{Spc}(\mathcal{K})$ corresponding to S. Suppose that $Z(S)$ is a Thomason subset and that $F\colon \mathcal{K} \to \mathcal{L}$ is an equivalence over $Z(S)$. As F is a tt-functor, the set $F(S)$ is a $\otimes$-multiplicative set of objects in $\mathcal{L}$ and $Z(F(S)) = \varphi^{-1}(Z(S))$. We are claiming that

$$\Psi(\mathcal{I}_{F(S)}) = (F_!)^{-1}(\mathcal{I}_{F(S)}) = \mathcal{I}_S.$$

Using Proposition 3.17, the inclusion $\mathcal{I}_S \subseteq (F_!)^{-1}(\mathcal{I}_{F(S)})$ is clear. For the reverse inclusion, suppose that $M \in \text{mod-}\mathcal{K}$ and $F_!(M) \in \mathcal{I}_{F(S)}$. This means that $0 = \mathrm{h}(F(a)) \otimes F_!(M) \cong F_!(\mathrm{h}(a) \otimes M)$ for some $a \in S$. We may assume that $\operatorname{supp}(a) = Z(S)$ by replacing a with $a_0 \otimes a$, where $a_0 \in S$ and $\operatorname{supp}(a_0) = Z(S)$ as in Remark 3.19. Then $F_!(\mathrm{h}(a) \otimes M) = 0$ in Mod-$\mathcal{L}$

forces $i^*(\mathrm{h}(a)\otimes M)=0$ in Mod-$\mathcal{K}_{Z(S)}$ by Lemma 5.7 applied to $Y=Z(S)$. Since $\mathrm{h}(a)\otimes M$ is $\mathrm{Ker}(i^*)$-local (Lemma 4.1), we get $\mathrm{h}(a)\otimes M=0$. In other words, $M\in\mathcal{I}_S$ as claimed.

To conclude this section, we apply our results to completion.

5.20. *Recollection.* Suppose that $\mathcal{K}$ comes from an ambient 'big' tt-category $\mathcal{T}$, as in (2.2). The *Bousfield completion* of $\mathcal{T}$ along a Thomason $Y\subseteq\mathrm{Spc}(\mathcal{K})$ is the subcategory $\mathcal{T}_Y^{\perp\perp}=\big((\mathcal{K}_Y)^\perp\big)^\perp$ of Y-complete objects in $\mathcal{T}$. This should not be mistaken for *the* completion $\widehat{\mathcal{T}}$ as a 'big' tt-category. See [BS26, Example 2.13 and Remark 5.4]. Only the dualizables of $\widehat{\mathcal{T}}$ should be expected to be found in $\mathcal{T}_Y^{\perp\perp}$ (and thus ultimately in $\mathcal{T}$!) and one *defines*

$$\widehat{\mathcal{T}}^{\mathrm{d}}\coloneqq(\mathcal{T}_Y^{\perp\perp})^{\mathrm{d}}.$$

The full $\widehat{\mathcal{T}}$ is the Ind-completion of $\widehat{\mathcal{T}}^{\mathrm{d}}$. Denote by $\mathbb{e}_Y\in\mathcal{T}$ the left idempotent of [BF11]. The localization $[\mathbb{e}_Y,-]\colon\mathcal{T}\to\mathcal{T}_Y^{\perp\perp}$ is a tensor functor and its restriction to dualizable objects is called the *Y-completion functor* $\mathrm{Compl}_Y\colon\mathcal{T}^{\mathrm{d}}\to\widehat{\mathcal{T}}^{\mathrm{d}}$. By [BS25, Theorem 4.6], it is an equivalence over Y. In light of this, the following, announced as Theorem 1.9, is immediate.

5.21. **Corollary.** *Keep notation as in Recollection 5.20. Then the Y-completion functor $\mathrm{Compl}_Y\colon\mathcal{T}^{\mathrm{d}}\to\widehat{\mathcal{T}}^{\mathrm{d}}$ induces an inclusion-preserving bijection between the Serre $\otimes$-ideals $\mathcal{I}\subseteq\text{mod-}\mathcal{T}^{\mathrm{d}}$ with $W^{\mathrm{h}}(\mathcal{I})\subseteq\pi^{-1}(Y)$ and the Serre $\otimes$-ideals $\hat{\mathcal{I}}\subseteq\text{mod-}\widehat{\mathcal{T}}^{\mathrm{d}}$ with $W^{\mathrm{h}}(\hat{\mathcal{I}})\subseteq\pi^{-1}(\varphi^{-1}(Y))$, such that we have an equivalence of the corresponding layers*

$$\frac{\text{Mod-}\mathcal{T}^{\mathrm{d}}}{\mathrm{Loc}(\mathcal{I})}\xrightarrow{\ \sim\ }\frac{\text{Mod-}\widehat{\mathcal{T}}^{\mathrm{d}}}{\mathrm{Loc}(\hat{\mathcal{I}})}.$$

In particular, if $\mathcal{I}\in\pi^{-1}(Y)$ is maximal, then $\hat{\mathcal{I}}\in\pi^{-1}(\varphi^{-1}(Y))$ is maximal as well and the above is an equivalence of homological residue fields. □

6. Residual regularity

Let $\mathcal{K}$ and $\mathcal{L}$ be tt-categories as in our Hypothesis 2.1 and $F\colon\mathcal{K}\to\mathcal{L}$ be a tt-functor that is an equivalence over a Thomason subset Y (Definition 5.1). We want to show that F preserves and detects residual regularity on Y. As usual, we write $\varphi\coloneqq\mathrm{Spc}(F)$ and $\varphi^{\mathrm{h}}\coloneqq\mathrm{Spc}^{\mathrm{h}}(F)$.

6.1. *Recollection.* For any prime $\otimes$-ideal $\mathcal{P}$ of $\mathcal{K}$, we define the *local category* at $\mathcal{P}$ to be $\mathcal{K}_{\mathcal{P}}\coloneqq(\mathcal{K}/\mathcal{P})^\natural$, which is again an essentially small, rigid and idempotent complete tt-category. By [Bal20b, Proposition 2.9] the functor $q_!\colon\text{mod-}\mathcal{K}\to\text{mod-}\mathcal{K}_{\mathcal{P}}$ induced by $q\colon\mathcal{K}\to\mathcal{K}_{\mathcal{P}}$ is a Gabriel quotient with kernel $\langle\mathrm{h}(\mathcal{P})\rangle$. It follows that $\mathcal{G}\mapsto(q_!)^{-1}(\mathcal{G})$ defines an inclusion-preserving one-to-one correspondence $\mathrm{Spc}^{\mathrm{h}}(\mathcal{K}_{\mathcal{P}})\xrightarrow{\sim}W^{\mathrm{h}}(\langle\mathrm{h}(\mathcal{P})\rangle)$ between homological primes $\mathcal{G}$ of $\mathcal{K}_{\mathcal{P}}$ and the homological primes $\mathcal{H}$ of $\mathcal{K}$ which contain the Serre $\otimes$-ideal $\langle\mathrm{h}(\mathcal{P})\rangle$. Moreover, if $\mathcal{G}$ corresponds to $\mathcal{H}$, then the respective homological residue fields are canonically equivalent in such a way that the following diagram is commutative; see [Bal20b, Corollary 3.6]:

$$\begin{array}{ccccc}
\mathcal{K} & \xhookrightarrow{h} & \text{Mod-}\mathcal{K} & \xrightarrow{Q_{\mathcal{H}}} & \text{Mod-}\mathcal{K}/\mathrm{Loc}(\mathcal{H}) \\
{\scriptstyle q}\downarrow & & \downarrow{\scriptstyle q_!} & & \downarrow\wr \\
\mathcal{K}_{\mathcal{P}} & \xhookrightarrow[h]{} & \text{Mod-}\mathcal{K}_{\mathcal{P}} & \xrightarrow[Q_{\mathcal{G}}]{} & \text{Mod-}\mathcal{K}_{\mathcal{P}}/\mathrm{Loc}(\mathcal{G}).
\end{array}$$

Since the injective hulls $\bar{E}_{\mathcal{H}}$ and $\bar{E}_{\mathcal{G}}$ are identified under the right-hand equivalence, we have $\kappa(\mathcal{H})\simeq q^*(\kappa(\mathcal{G}))$ in Mod-$\mathcal{K}$ and $q_!(\kappa(\mathcal{H}))\simeq\kappa(\mathcal{G})$ in Mod-$\mathcal{K}_{\mathcal{P}}$. In particular, $\kappa(\mathcal{H})$ is a $\mathrm{Ker}(q_!)$-local object and it makes sense to speak of $\kappa(\mathcal{H})$ as an object in Mod-$\mathcal{K}_{\mathcal{P}}$.

The following definition is essentially [VR26, Definition 2.1].

6.2. **Definition.** The tt-category $\mathcal{K}$ is called *residually regular at the point* $\mathcal{P} \in \operatorname{Spc}(\mathcal{K})$ if there exists a homological prime $\mathcal{H}$ in $\pi^{-1}(\{\mathcal{P}\})$ and an object a in the local category $\mathcal{K}_{\mathcal{P}} = (\mathcal{K}/\mathcal{P})^{\natural}$ whose image under Yoneda $\mathrm{h}\colon \mathcal{K}_{\mathcal{P}} \hookrightarrow \text{Mod-}\mathcal{K}_{\mathcal{P}}$ is isomorphic to $\kappa(\mathcal{H})$.

6.3. *Remark.* In [VR26], the tt-category $\mathcal{K}$ is assumed to be the small part of a 'big' tt-category as in (2.2). This allows us to produce $\kappa(\mathcal{H})$ in $\mathcal{T}$ by Brown Representability (see [Kra00, Corollary 1.9]). Since an object $X \in \mathcal{T}$ becomes isomorphic to a compact $a \in \mathcal{T}^{\mathrm{c}}$ in Mod-$\mathcal{T}^{\mathrm{c}}$ if and only $X \simeq a$ in $\mathcal{T}$, Definition 6.2 coincides with [VR26, Definition 2.1] in this case.

6.4. *Remark.* From Recollection 6.1, it follows that $\mathcal{K}$ is residually regular at the point $\mathcal{P} \in \operatorname{Spc}(\mathcal{K})$ if and only the local category $\mathcal{K}_{\mathcal{P}}$ is residually regular at its unique closed point (0).

6.5. **Proposition.** *Suppose that $\mathcal{K}$ is a local tt-category. Then $\mathcal{K}$ is residually regular at the closed point (0) if and only if there exists a homological prime in $\mathcal{H} \in \pi^{-1}\{(0)\}$ such that $\kappa(\mathcal{H})$ is finitely presented in* Mod-$\mathcal{K}$. *In that case, the unique object $a \in \mathcal{K}$ such that $\mathrm{h}(a) \simeq \kappa(\mathcal{H})$ has support exactly the closed point,* $\operatorname{supp}(a) = \{(0)\}$, *and* $\pi^{-1}(\{(0)\}) = \{\mathcal{H}\}$.

Proof. If $\kappa(\mathcal{H})$ is finitely presented then it is an injective object in mod-$\mathcal{K}$. Hence it must come from a unique $a \in \mathcal{K}$, that is, $\mathrm{h}(a) \simeq \kappa(\mathcal{H})$. For every $\mathcal{G} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$ the object $\kappa(\mathcal{G}) \otimes \mathrm{h}(a) \simeq \kappa(\mathcal{G}) \otimes \kappa(\mathcal{H})$ is zero if and only if $\mathcal{G} \neq \mathcal{H}$ (see [Bal20b, Proposition 5.3]). Hence $\pi^{-1}(\operatorname{supp}(a)) = \{\mathcal{G} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \,|\, \kappa(\mathcal{G}) \otimes \mathrm{h}(a) \neq 0\} = \{\mathcal{H}\}$. By surjectivity of π, we get $\operatorname{supp}(a) = \pi(\pi^{-1}(\operatorname{supp}(a))) = \pi(\{\mathcal{H}\}) = \{(0)\}$ and $\pi^{-1}(\{(0)\}) = \pi^{-1}(\operatorname{supp}(a)) = \{\mathcal{H}\}$. □

6.6. **Lemma.** *Suppose that $\mathcal{K}$ is a local tt-category and let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a nonempty Thomason subset and $\mathcal{H}$ a homological prime in $\pi^{-1}\{(0)\}$. Then, the residue object $\kappa(\mathcal{H})$ is finitely presented in* Mod-$\mathcal{K}$ *if and only if $i^*(\kappa(\mathcal{H}))$ is finitely presented in* Mod-$\mathcal{K}_Y$.

Proof. First, suppose that $\kappa(\mathcal{H})$ is finitely presented in Mod-$\mathcal{K}$. By Proposition 6.5, there exists an object $a \in \mathcal{K}$ with $\operatorname{supp}(a) = \{(0)\}$ and such that $\kappa(\mathcal{H}) \simeq \mathrm{h}(a)$. Since Y is a nonempty Thomason subset, it must contain the closed point (0) and therefore $a \in \mathcal{K}_Y$. It follows that $i^*(\kappa(\mathcal{H})) \cong i^*(\mathrm{h}(a)) = \operatorname{Hom}_{\mathcal{K}_Y}(-, a)$ is finitely presented in Mod-$\mathcal{K}_Y$.

Conversely, suppose that $i^*(\kappa(\mathcal{H}))$ is finitely presented in Mod-$\mathcal{K}_Y$. Since i^* has an exact left adjoint $i_!$ (see Recollection 2.6), it preserves injective objects. So, $i^*(\kappa(\mathcal{H}))$ is in fact an injective object in mod-$\mathcal{K}_Y$, therefore $i^*(\kappa(\mathcal{H})) \simeq \mathrm{h}(a)$ for some $a \in \mathcal{K}_Y$. Note that $\kappa(\mathcal{H})$ is a $\operatorname{Ker}(i^*)$-local object since it is $\operatorname{Loc}(\mathcal{H})$-local, and we have $\operatorname{Ker}(i^*) \subseteq \operatorname{Loc}(\mathcal{H})$ by Theorem 4.6 (because $\pi(\mathcal{H}) = (0) \in Y$). Therefore $\kappa(\mathcal{H}) \cong i_! i^*(\kappa(\mathcal{H})) \simeq i_!(\mathrm{h}(a)) \cong \mathrm{h}(i(a)) \in \text{mod-}\mathcal{K}$. □

6.7. **Theorem.** *Let $F\colon \mathcal{K} \to \mathcal{L}$ be a tt-functor. Let $Y \subseteq \operatorname{Spc}(\mathcal{K})$ be a nonempty Thomason subset such that F is an equivalence over Y. Let $\mathcal{Q} \in \varphi^{-1}(Y)$ and set $\mathcal{P} = \varphi(\mathcal{Q})$. Then $\mathcal{L}$ is residually regular at $\mathcal{Q}$ if and only if $\mathcal{K}$ is residually regular at $\mathcal{P}$.*

Proof. Since $\mathcal{P} = F^{-1}(\mathcal{Q})$ we get a conservative localization $\bar{F}\colon \mathcal{K}_{\mathcal{P}} \to \mathcal{L}_{\mathcal{Q}}$ as in Proposition 5.4 applied to $U = \operatorname{Supp}(\mathcal{P})^{\complement}$ and $V = \operatorname{Supp}(\mathcal{Q})^{\complement}$. By that proposition, we can assume that $\mathcal{K}$ and $\mathcal{L}$ are local tt-categories and $\mathcal{P}$ and $\mathcal{Q}$ are their respective closed points. Since residual regularity is a local property (see Remark 6.4), this suffices to prove the statement.

Corollary 5.14 implies that φ^{h} restricts to a bijection $\pi^{-1}(\{\mathcal{Q}\}) \xrightarrow{\sim} \pi^{-1}(\{\mathcal{P}\})$. Consider a matching pair of points $\mathcal{G} \in \pi^{-1}(\{\mathcal{Q}\})$ and $\mathcal{H} = \varphi^{\mathrm{h}}(\mathcal{G})$. Writing as before $i\colon \mathcal{K}_Y \hookrightarrow \mathcal{K}$ and $j\colon \mathcal{L}_{Y'} \hookrightarrow \mathcal{L}$ for the inclusions we have $j \circ F_Y = F \circ i$ and therefore $(F_Y)^* j^*(\kappa(\mathcal{G})) \cong i^* F^*(\kappa(\mathcal{G})) \simeq i^*(\kappa(\mathcal{H}))$, where the last isomorphism uses Corollary 5.18. Since $(F_Y)^*$:

Mod-$\mathcal{L}_{Y'} \xrightarrow{\sim}$ Mod-$\mathcal{K}_Y$ is an equivalence, it follows that $j^*(\kappa(\mathcal{G}))$ is finitely presented if and only if $i^*(\kappa(\mathcal{H}))$ is. We can now translate these properties into the residual regularity of $\mathcal{K}$ at $\mathcal{P} = (0)$, respectively of $\mathcal{L}$ at $\mathcal{Q} = (0)$ thanks to Lemma 6.6 and Proposition 6.5. □

Theorem 6.7 applies in particular to the completion as in Recollection 5.20.

6.8. **Corollary.** *Let $\mathcal{T}$ be a rigidly-compactly generated tt-category, and let $Y \subseteq \operatorname{Spc}(\mathcal{T}^{\mathrm{d}})$ be a nonempty Thomason subset. Let $\hat{\mathcal{P}} \in \varphi^{-1}(Y)$ and set $\mathcal{P} = \varphi(\hat{\mathcal{P}})$. Then $\widehat{\mathcal{T}}^{\mathrm{d}}$ is residually regular at $\hat{\mathcal{P}}$ if and only if $\mathcal{T}^{\mathrm{d}}$ is residually regular at $\mathcal{P}$.* □

7. Example: Commutative rings

Let R be a commutative ring. In this section, we consider layers of $\mathcal{K} \coloneqq \mathrm{D}^{\mathrm{perf}}(R)$. In particular, we study the interplay of ideals $I \subseteq R$ and Serre $\otimes$-ideals $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$.

7.1. *Recollection.* The tt-category $\mathrm{D}^{\mathrm{perf}}(R)$ has the steel property $\pi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{K}) \xrightarrow{\sim} \operatorname{Spc}(\mathcal{K})$ by [Bal20b, Corollary 5.11]. Its tt-spectrum identifies with the Zariski spectrum of the ring R via the comparison map $\operatorname{comp}\colon \operatorname{Spc}(\mathcal{K}) \xrightarrow{\sim} \operatorname{Spec}(R)$ defined by $\operatorname{comp}(\mathcal{P}) = \big\{r \in R \,\big|\, \operatorname{cone}(r) \notin \mathcal{P}\big\}$. See [Bal10]. We sometimes drop the homeomorphisms π and comp for readability. The comparison map satisfies $\operatorname{comp}^{-1}(V(r)) = \operatorname{supp}(\operatorname{cone}(r))$ for every $r \in R$ and consequently

$$\operatorname{comp}^{-1}(V(r_1, \dots, r_n)) = \operatorname{supp}(\operatorname{kos}(\underline{r})) \qquad \text{where} \qquad \operatorname{kos}(\underline{r}) = \otimes_{i=1}^{n} \operatorname{cone}(r_i) \tag{7.2}$$

is the Koszul complex associated to the sequence $\underline{r} = (r_1, \dots, r_n)$.

7.3. *Construction.* Given an ideal $I \subseteq R$, let $S_I = \big\{ \operatorname{kos}(\underline{r}) \,\big|\, \underline{r} \text{ in } I \big\}$ be the $\otimes$-multiplicative subset of $\mathcal{K} = \mathrm{D}^{\mathrm{perf}}(R)$ generated by $\big\{ \operatorname{cone}(r) \,\big|\, r \in I \big\}$. Denote the Serre $\otimes$-ideal $\mathcal{I}_{S_I}$ of Definition 3.15 by $\sigma(I)$. The reader can verify that $\ker(\mathrm{h}(\operatorname{coev}_{\operatorname{cone}(r)})) \cong \operatorname{im}(\mathrm{h}(r))$ for every element $r \in R$, so the Serre $\otimes$-ideal $\sigma(I)$ is generated by $\big\{ \operatorname{im}(\mathrm{h}(r)) \,\big|\, r \in I \big\}$ in mod-$\mathcal{K}$. By Proposition 3.17, we also have the following description

$$\sigma(I) = \big\{ M \in \text{mod-}\mathcal{K} \,\big|\, \mathrm{h}(\operatorname{kos}(\underline{r})) \otimes M = 0 \text{ for some } \underline{r} = (r_1, \dots, r_n) \text{ in } I \big\}.$$

In the other direction, given a Serre $\otimes$-ideal $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$, define an ideal of R

$$\tau(\mathcal{I}) \coloneqq \big\{ r \in R \,\big|\, \operatorname{im}(\mathrm{h}(r)) \in \mathcal{I} \big\} = \ker\big(\operatorname{End}_{\mathcal{A}}(\mathbb{1}) \xrightarrow{Q_{\mathcal{I}}} \operatorname{End}_{\bar{\mathcal{A}}_{\mathcal{I}}}(\mathbb{1}) \big)$$

using that $R \cong \operatorname{End}_{\mathcal{K}}(\mathbb{1}) \cong \operatorname{End}_{\mathcal{A}}(\mathbb{1})$ and the exact $\otimes$-functor $Q_{\mathcal{I}}\colon \mathcal{A} = \text{Mod-}\mathcal{K} \twoheadrightarrow \bar{\mathcal{A}}_{\mathcal{I}}$.

7.4. **Proposition.** *The functions σ and τ of Construction 7.3 are compatible with vanishing loci in the following sense.*

(a) *For every ideal $I \subseteq R$, we have $\operatorname{comp}\pi(W^{\mathrm{h}}(\sigma(I))) = V(I)$ in $\operatorname{Spec}(R)$. Omitting the homeomorphisms for readability, this amounts to*

$$W(\sigma(I)) = V(I).$$

In particular, the $\sigma(I)$-layer is closed.

(b) *For every Serre $\otimes$-ideal $\mathcal{I} \subseteq \text{mod-}\mathcal{K}$, the closure of $\operatorname{comp}\pi(W^{\mathrm{h}}(\mathcal{I}))$ in $\operatorname{Spec}(R)$ is $V(\tau(\mathcal{I}))$. Omitting the homeomorphisms again, this reads*

$$\overline{W(\mathcal{I})} = V(\tau(\mathcal{I})).$$

In particular, we have $W(\mathcal{I}) \subseteq V(\tau(\mathcal{I}))$, with equality if and only if the $\mathcal{I}$-layer is closed.

Proof. (a): We use the characterization $\sigma(I) = \mathcal{I}_S$, where $S = \{\operatorname{kos}(\underline{r}) \mid \underline{r} \text{ in } I\}$. By Corollary 3.16, the spectrum $W(\sigma(I)) = W(\mathcal{I}_S)$ is

$$\bigcap_{a\in S} \operatorname{supp}(a) = \bigcap_{\underline{r} \text{ in } I} \operatorname{supp}(\operatorname{kos}(\underline{r})) \overset{(7.2)}{=} \bigcap_{\underline{r} \text{ in } I} \operatorname{comp}^{-1}(V(\underline{r})) = \operatorname{comp}^{-1} \bigcap_{\underline{r} \text{ in } I} V(\underline{r}) = \operatorname{comp}^{-1} V(I).$$

(b): *Claim: For $r \in R$, we have $\operatorname{comp} W(\mathcal{I}) \subseteq V(r)$ if and only if $r \in \sqrt{\tau(\mathcal{I})}$.* Suppose first that $\operatorname{comp} W(\mathcal{I}) \subseteq V(r)$. This means that for every $\mathcal{H} \supseteq \mathcal{I}$, we have $r \in \operatorname{comp} \pi(\mathcal{H})$, that is, $\overline{\operatorname{cone}(r)} \neq 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$. Since $r \otimes \operatorname{cone}(r) = 0$, we must have $\overline{r} = 0$ in $\bar{\mathcal{A}}_{\mathcal{H}}$ for every $\mathcal{H} \supseteq \mathcal{I}$. By Recollection 3.3, this forces $r\colon \mathbb{1} \to \mathbb{1}$ to become $\otimes$-nilpotent in $\bar{\mathcal{A}}_{\mathcal{I}}$, meaning $r \in \sqrt{\tau(\mathcal{I})}$. Conversely, if $r^n \in \tau(\mathcal{I})$ for $n \geq 1$, then $r^n = 0$ in $\bar{\mathcal{A}}_{\mathcal{I}}$ and a fortiori in $\bar{\mathcal{A}}_{\mathcal{H}}$ for every $\mathcal{H} \in W^{\mathrm{h}}(\mathcal{I})$. This means that $\operatorname{cone}(r^n)$ goes to a non-zero object in $\bar{\mathcal{A}}_{\mathcal{H}}$ and thus $r^n \in \operatorname{comp} \pi(\mathcal{H})$, forcing $r \in \operatorname{comp} \pi(\mathcal{H})$ by primality. This proves the Claim. Since $\{V(r)\}_{r\in R}$ forms a closed basis of the Zariski topology, the Claim implies that the closure of $\operatorname{comp} W(\mathcal{I})$ is the intersection of all the $V(r)$ for $r \in \sqrt{\tau(\mathcal{I})}$. This intersection is $V(\sqrt{\tau(\mathcal{I})}) = V(\tau(\mathcal{I}))$. □

The next proposition shows that the collection of closed layers of $\mathrm{D}^{\mathrm{perf}}(R)$ carries at least as much information as the collection of quotients of R.

7.5. **Proposition.** *For every ideal $I \subseteq R$, we have $\tau(\sigma(I)) = I$. So we have an injection*

$$\sigma\colon \{\textit{ideals of } R\} \to \{\textit{closed layers of } \mathrm{D}^{\mathrm{perf}}(R)\}.$$

Proof. We have $\tau(\sigma(I)) = \{x \in R \mid \operatorname{im}(\mathrm{h}(x)) \in \sigma(I)\}$. Certainly, $I \subseteq \tau(\sigma(I))$, since $\operatorname{cone}(x) \otimes x = 0$ for any $x \in I$. For the other inclusion, if $x \in R$ satisfies $\mathrm{h}(\operatorname{kos}(\underline{r})) \otimes \operatorname{im}(\mathrm{h}(x)) = 0$ for $\underline{r} = (r_1, \dots, r_n)$ in I, then $\operatorname{kos}(\underline{r}) \otimes x = 0$. So multiplication by x is zero on the homology of $\operatorname{kos}(\underline{r})$, including $H_0(\operatorname{kos}(\underline{r})) = R/\langle r_1, \dots, r_n\rangle$. This implies $x \in \langle r_1, \dots, r_n\rangle \subseteq I$. □

7.6. *Remark.* In general, one cannot expect σ to be a bijection. For example, let $R = \mathbb{Z}$. The only ideal I with $V(I) = \operatorname{Spec} R$ is $I = (0)$. However, there are many Serre $\otimes$-ideals $\mathcal{I}$ other than $\sigma((0)) = (0)$ for which $W^{\mathrm{h}}(\mathcal{I}) = \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. For example, let c be the complex $\mathbb{Z} \xrightarrow{4} \mathbb{Z}$, and consider the map $f = (c \xrightarrow{2} c)$. Then f vanishes in every residue field, hence $\mathcal{I} := \langle \operatorname{im}(\mathrm{h}(f))\rangle$ is contained in every $\mathcal{H} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{K})$. So $W(\mathcal{I}) = \operatorname{Spec}(\mathbb{Z})$ and $\tau(\mathcal{I})$ must be 0, but $\mathcal{I} \neq 0$.

7.7. *Remark.* Given an ideal $I \subseteq R$, there is another Serre $\otimes$-ideal of mod-$\mathcal{K}$ associated to I, namely the kernel of the functor $F_!\colon$ mod-$\mathcal{K} \to$ mod-$\mathrm{D}^{\mathrm{perf}}(R/I)$ induced by the ring map $R \twoheadrightarrow R/I$. This kernel contains $\sigma(I)$ but can be larger in general. For example, let k be a field of characteristic 0, let $R = k[t]/(t^2)$, and let f be the following morphism of complexes:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & R & \xrightarrow{t} & R & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow t & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & R & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

Letting $I = (t)$, we see that $\operatorname{im}(\mathrm{h}(f)) \in \operatorname{Ker}(F_!)$. However, $f \otimes \operatorname{cone}(t) \neq 0$. In fact, one can verify directly that no nonzero compact object $c \in \mathrm{D}^{\mathrm{perf}}(R)$ annihilates f (using that $\operatorname{char}(k) = 0$). See [AM15] for details on the category $\mathrm{D}^{\mathrm{perf}}(k[t]/(t^2))$.

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UCLA Mathematics Department, Los Angeles, CA 90095-1555, USA

Email address: balmer@math.ucla.edu, rkommerell@math.ucla.edu, emmyvr@math.ucla.edu